

# Chapter 3: Strategies

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## 3.1 Introduction

This chapter tries to lay out the big picture of AI safety strategy to mitigate the risks explored previously.

AI capabilities advance very rapidly, the strategies designed to ensure safety must also evolve. The first version of this document was written in summer of 2024, this version includes the update during the summer of 2025. Through the course of this chapter, we aim to provide a structured overview of the thinking and ongoing work in AI safety strategy as of 2025. We acknowledge both established methods and emerging research directions.

We have categorized mitigations around preventing misuse of AI, safety mitigations for AGI and ASI, and finally socio-technical approaches that help mitigate concerns more generally across all categories. Even though we have chosen a decomposition for sake of explanation, we advocate for a comprehensive approach that combines many of these strategies instead of pursuing just a few in isolation. Finally we have a combined strategies section, where we attempt to outline one potential way that this combination could look to create a layered defense-in-depth framework.

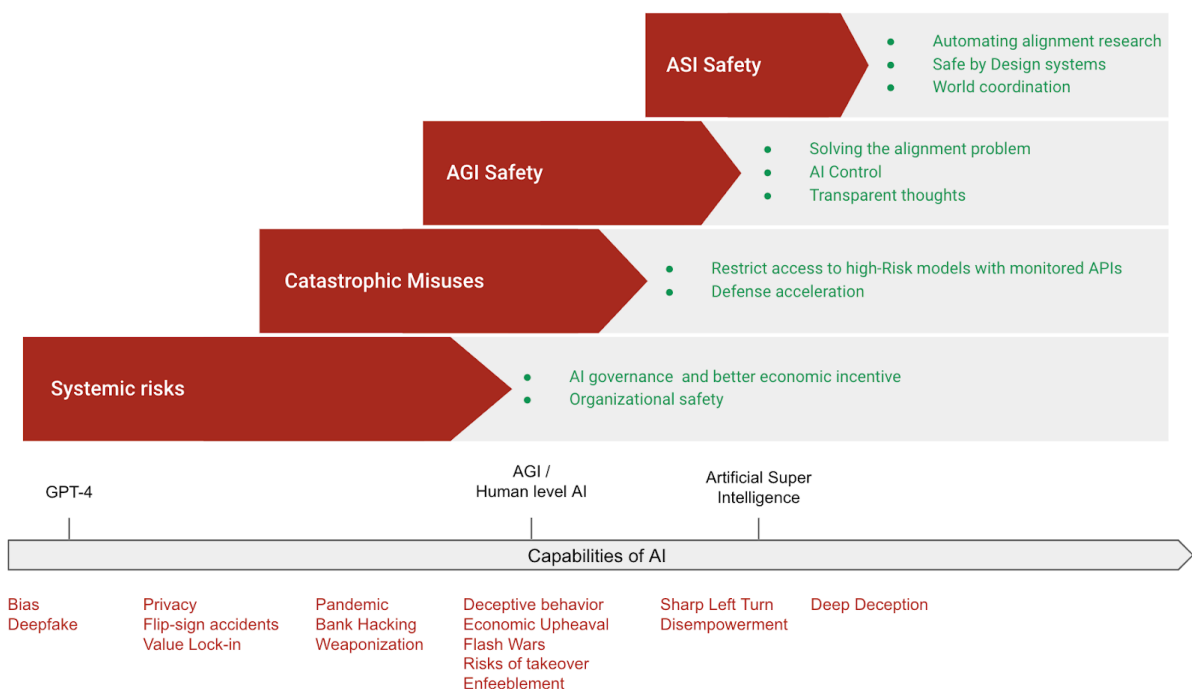


Figure 3.1: Tentative diagram summarizing the main high-level approaches to make AI development safe.

## Beyond the scope of this chapter

### OPTIONAL NOTE

While this chapter focuses on strategies directly related to preventing large-scale negative outcomes from AI misuse, misalignment, or uncontrolled development, several related topics are necessarily placed beyond its primary scope:

- **AI-generated misinformation:** The proliferation of AI-driven misinformation, including deepfakes and biased content generation. Strategies to combat this, such as robust detection systems, watermarking, and responsible AI principles, are mostly beyond the scope of the chapter. These often fall under the umbrella of content moderation, media literacy, and platform governance, distinct from the core technical alignment and control strategies discussed in this chapter.
- **Privacy:** AI systems often process vast amounts of data, amplifying existing concerns about data privacy.
- **Security:** Standard security practices, such as encryption, access control, data classification, threat monitoring, and anonymization, are prerequisites for safe AI deployment. Although robust security is vital for measures such as protecting model weights, these standard practices are distinct from the novel safety strategies required to address risks like model misalignment or capability misuse.
- **Discrimination and toxicity:** While biased or toxic outputs constitute a safety concern, this chapter concentrates on strategies aimed at preventing catastrophic failures.
- **Digital mind welfare and rights:** We don't know if AIs should be considered as moral patients. This is a distinct ethical domain concerning our obligations to AI, rather than ensuring safety from AI.
- **Errors due to lack of capability:** While AI system failures due to a lack of capability or robustness are a source of risk ([AISI, 2025](#)), the strategies discussed in this chapter aim to mitigate risks arising from both insufficient robustness and potentially high (but misaligned or misused) capabilities. The solutions to this type of risk are the same as those for other industries: testing, iteration, and enhancing the system's capabilities.

The scope chosen here reflects a common focus within certain parts of the AI safety community on existential or large-scale catastrophic risks arising from powerful, potentially agentic AI systems.

## 3.2 Definitions

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How we define problems directly impacts which strategies we pursue in solving that problem. In a new and evolving field like AI safety, clearly defined terms are essential for effective communication and research. Ambiguity leads to miscommunication, hinders collaboration, obscures disagreements, and facilitates safety washing ([Ren et al., 2024](#); [Lizka, 2023](#)). The terms we use reflect our assumptions about the nature of the problems we're trying to solve and shape the solutions we develop. Terms like "alignment" and "safety" are used with varying meanings, reflecting different underlying assumptions about the nature of the problem and the research goals. The goal of this section is to explain different perspectives on these words, what specific safety strategies aim to achieve, and establish how our text will utilize them.

### 3.2.1 AI Safety

---

#### AI SAFETY

##### DEFINITION 3.1

*Ensuring that AI systems do not inadvertently or deliberately cause harm or danger to humans or the environment, through research that identifies causes of unintended AI behavior and develops tools for safe and reliable operation.*

**AI safety ensures AI systems do not cause harm to humans or the environment.** It encompasses the broadest range of research and engineering practices focused on preventing harmful outcomes from AI systems. While alignment focuses on aspects such as an AI's goals and intentions, safety addresses a broader range of concerns ([Rudner et al., 2021](#)). It is concerned with ensuring that AI systems do not inadvertently or deliberately cause harm or danger to humans or the environment. AI safety research seeks to identify the causes of unintended AI behavior and develop tools for ensuring safe and reliable operation. It can include technical subfields like robustness (ensuring reliable performance, including against adversarial attacks), monitoring (observing AI behavior), and capability control (limiting potentially dangerous abilities).

### 3.2.2 AI Alignment

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#### AI ALIGNMENT

DEFINITION 3.2

(Christiano, 2024)

*The problem of building machines that faithfully try to do what we want them to do (or what we ought to want them to do).*

**AI alignment aims to ensure AI systems act in accordance with human intentions and values.** Alignment is a subset of safety that focuses specifically on the technical problem of ensuring AI objectives align with human intentions and values. Theoretically, a system could be aligned but unsafe (e.g., competently pursuing the wrong goal due to misspecification) or safe but unaligned (e.g., constrained by control mechanisms despite misaligned objectives). While this sounds straightforward, the precise scope varies significantly across research communities. We already saw a brief definition of alignment in the previous chapter, but this section offers a more nuanced perspective on the various definitions that we could potentially work with.

#### What Do We Mean by 'Alignment'?

OPTIONAL NOTE

**Broader definitions of alignment encompass the entire challenge of creating beneficial AI outcomes.** These approaches focus on ensuring that AI systems understand and properly implement human preferences (Christiano, 2018), address complex value learning challenges (Dewey, 2011), and incorporate robustness aspects, such as resistance to jailbreaking (Jonker et al., 2024). This comprehensive view treats alignment as encompassing both the system's intent and its ability to understand human values – essentially addressing the full spectrum of what makes an AI system behave in ways that humans would approve of<sup>a</sup>.

**Narrower definitions of alignment focus specifically on the AI's motivation and intent,**

**independent of outcomes.** Some definitions are much more narrow and focus specifically on the AI's motivation – *"An AI (A) is trying to do what a human operator (H) wants it to do"* (Christiano, 2018). This emphasizes the AI's motivation rather than its competence or knowledge. Under this definition, an intent-aligned AI might still fail due to misunderstanding the operator's wishes or lacking knowledge about the world, but it is fundamentally trying to be helpful. Proponents argue this narrow focus isolates the core technical challenge of getting AI systems to adopt human goals, separate from broader issues like value clarification or capability robustness. That is, as long as the agent "means well", it is aligned, even if errors in its assumptions about the user's preferences or about the world at large, lead it to actions that are bad for the user.

The choice of definition reflects underlying assumptions about AI risk and promising solutions. Focusing narrowly on intent alignment prioritizes research on inner/outer alignment problems, whereas broader views incorporate value learning or robustness research more centrally. These different approaches lead to other research priorities and safety strategies.

**Applying concepts like "trying," "wanting," or "intent" to AI systems is non-trivial.** When we train AI systems, we specify an optimization objective (like maximizing a reward function), but this doesn't necessarily translate to the system "intending" to pursue that objective in a human-like way. As we explained in the previous chapter, specification failures occur when what we specify doesn't capture what we actually want (a well-intended pursuit of a bad goal). But solving this is insufficient; it could also pursue completely different goals altogether. As an analogy, think about how evolution "optimized" humans for genetic fitness (optimization objective), yet humans developed other goals (like art appreciation or contraception) that don't maximize reproductive fitness. Similarly, AI systems optimized for specific objectives may develop internal "goals" that don't directly align with those objectives, especially as they become more capable.

**"Aligned to whom?" remains a fundamental question with no consensus answer.** Should AI systems align to the immediate operator (Christiano, 2018), the system designer (Gil, 2023), a specific group of humans, humanity as a whole (Miller, 2022), objective ethical principles, or the operator's hypothetical informed preferences? There are no agreed-upon answers to any of these questions, just many different perspectives, each with its own set of pros and cons.

We have tried to summarize some of the positions in the appendix.

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<sup>a</sup>While AI alignment does not necessarily encompass all systemic risks and misuse, there is some overlap. Some alignment techniques could help mitigate specific misuse scenarios—for instance, alignment methods could ensure that models refuse to cooperate with users intending to use AI for



harmful purposes, such as bioterrorism. Similarly, from a systemic risk perspective, a well-aligned AI might recognize and refuse to participate in problematic processes embedded within systems, such as financial markets. However, challenges remain, as malicious actors might attempt to circumvent these protections through targeted fine-tuning of models for harmful purposes, and in this case, even a perfectly aligned model wouldn't be able to resist.

### 3.2.3 AI Ethics

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#### AI ETHICS

DEFINITION 3.3

(Huang et al., 2023)

*The study and application of moral principles to AI development and deployment, addressing questions of fairness, transparency, accountability, privacy, autonomy, and other human values that AI systems should respect or promote.*

**AI ethics is the field that examines the moral principles and societal implications of AI systems.** It addresses the ethical considerations of potential societal upheavals resulting from AI advancements and the moral frameworks necessary to navigate these changes. The core of AI ethics lies in ensuring that AI developments are aligned with human dignity, fairness, and societal well-being, through a deep understanding of their broader societal impact. Research in AI ethics would encompass, for example, privacy norms, identifying and mitigating bias in systems (Huang et al., 2022; Harvard, 2025; Khan et al., 2022).

Ethics complements technical safety approaches by providing normative guidance on what constitutes beneficial AI outcomes. Alignment focuses on ensuring AI systems pursue intended objectives, research in ethics focuses on which objectives are worth pursuing (Huang et al., 2023; LaCroix & Luccioni, 2022). AI ethics might also include discussions of digital rights and potentially even the rights of digital minds, and AIs in the future.

This chapter focuses primarily on safety frameworks as they inform technical safety and governance strategies rather than exploring ethics, meta-ethics or digital rights.

### 3.2.4 AI Control

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#### AI CONTROL

DEFINITION 3.4

(Greenblatt et al., 2024)

*The technical and procedural measures designed to prevent AI systems from causing unacceptable outcomes, even if these systems actively attempt to subvert safety measures. Control focuses on maintaining human oversight regardless of whether the AI's objectives align with human intentions.*

**AI control ensures systems remain under human authority despite potential misalignment.** AI control implements mechanisms to ensure AI systems remain under human direction, even when they might act against our interests. Unlike alignment approaches that focus on giving AI systems the right goals, control addresses what happens if those goals diverge from human intentions (Greenblatt et al., 2024).

**Control and alignment work as complementary safety approaches.** While alignment aims to prevent preference divergence by designing systems with the right objectives, control creates layers of security that function even when alignment fails. Control measures include monitoring AI actions, restricting system capabilities, human auditing processes, and mechanisms to terminate AI systems when necessary (Greenblatt et al., 2023). Some researchers argue that even if alignment is needed for superintelligence-level AIs, control through monitoring may be a working strategy for less capable systems (Greenblatt et al., 2024). Ideally, an AGI would be aligned and controllable, meaning it would have the right goals and be subject to human oversight and intervention if something goes wrong.

The control line of AI safety work is discussed in much more detail in our chapter on AI evaluations.

## 3.3 Misuse Prevention Strategies

Strategies to prevent misuse often focus on controlling access to dangerous capabilities or implementing technical safeguards to limit harmful applications.

### 3.3.1 Access Controls

#### 3.3.1.1 External Controls

**Access control strategies directly address the inherent tension between open-sourcing benefits and misuse risks.** The AI industry has moved beyond binary discussions of "release" or "don't release"; instead, practitioners think in terms of a continuous gradient of access to models ([Kapoor et al., 2024](#)). The question of who gets access to a model sits on a range from fully closed (internal use only) to fully open (publicly available model weights with no restrictions).

#### OPEN SOURCE AI

DEFINITION 3.5

([Open Source Initiative, 2025](#))

*An Open Source AI is an AI system made available under terms and in a way that grants the freedoms to:*

- *Use the system for any purpose without having to ask for permission.*
- *Study how the system works and inspect its components.*
- *Modify the system for any purpose, including to change its output.*
- *Share the system for others to use with or without modifications, for any purpose.*

**Among these various access options, API-based deployment represents one of the most commonly used strategic middle grounds.** When we discuss access controls in this section, we're primarily talking about mechanisms that create a controlled gateway to AI capabilities—most commonly through API-based deployment, where most of the model (code, weights, and data) remain fully closed, but access to model capabilities is

partially open. In this arrangement, developers retain control over how their models are accessed and used. API-based controls maintain developer oversight, allowing continuous monitoring, updating of safety measures, and the ability to revoke access when necessary (Seger et al., 2023).

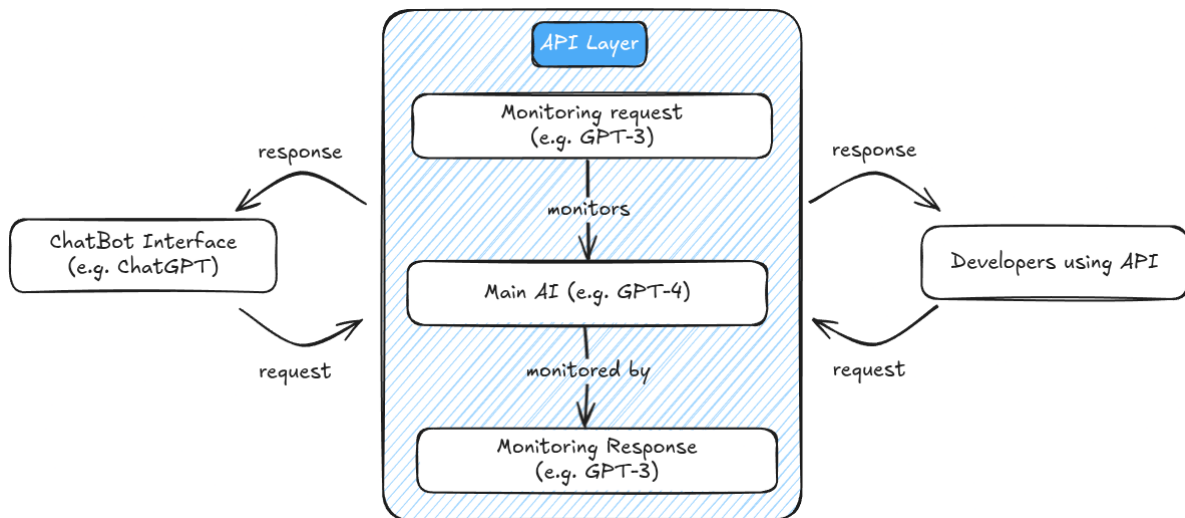


Figure 3.2: This is a simplified diagram to illustrate conceptually how an API would work. This is not how OpenAI's API works. It is for illustration purposes only.

**API-based deployment establishes a protective layer between users and model capabilities.** Instead of downloading model code or weights, users interact with the model by sending requests to a server where the model runs, receiving only the generated outputs in return. This architecture enables developers to implement various safety mechanisms:

- **Input/Output Filtering:** Screening prompts for harmful content and filtering generated responses according to safety policies. For example, filters can detect and block attempted generation of CSAM or instructions for building weapons. This approach directly counters misuses like generating illegal content or dangerous instructions.
- **Rate Limiting:** Preventing large-scale misuse through usage caps. By restricting the volume of requests, these controls mitigate risks of automated abuse like generating thousands of deepfakes or spam messages (Liang et al., 2022).
- **Usage Monitoring:** Beyond controlling request volume, usage monitoring enables identity and background checks for malicious users (similar to know your customer (KYC) laws). For example, this allows regulatory oversight, prevents repeated attempts to circumvent safety filters, and also enables deeper access to highly trusted users (Egan & Heim, 2023).
- **Usage Restrictions:** Enforcing terms of service that prohibit harmful applications. Companies can restrict high-risk applications like bioweapon research or autonomous

cyber operations through legal agreements backed by technical monitoring (Anderljung et al., 2023). When violations are detected, access can be revoked.

- **On-the-fly Updates:** Rapidly deploying improvements to safety systems without user action. Unlike open-sourced models where unsafe versions persist indefinitely, API-based models can be continually improved to address newly discovered vulnerabilities (Weidinger et al., 2023). This helps counter novel attack vectors like jailbreaking techniques.

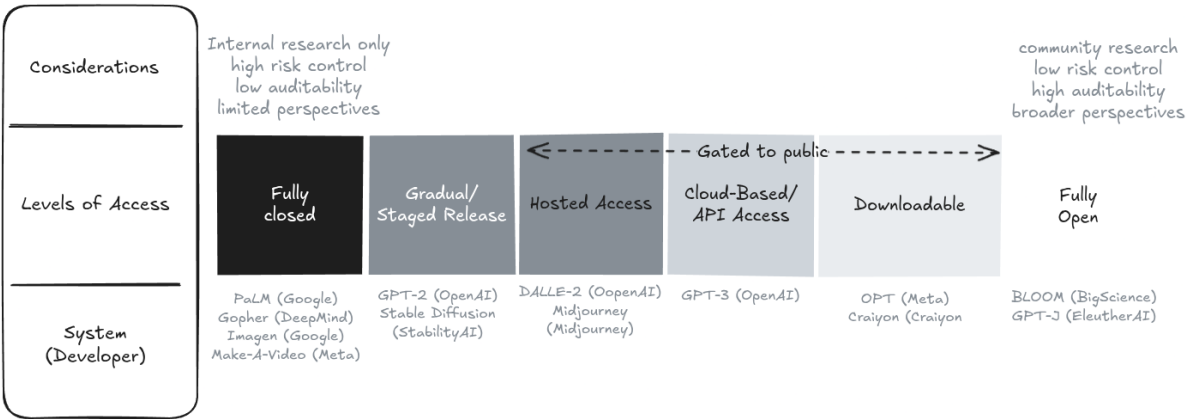


Figure 3.3: The gradient of access to AI models to the external public. Model release exists on a spectrum, from fully closed systems accessible only internally, to staged releases, API access, downloadable weights with restrictions, and fully open-source releases. API-based deployment represents an intermediate point on this gradient (Seeger et al., 2023).

## Different components of a model can exist at different points on the access spectrum

OPTIONAL NOTE

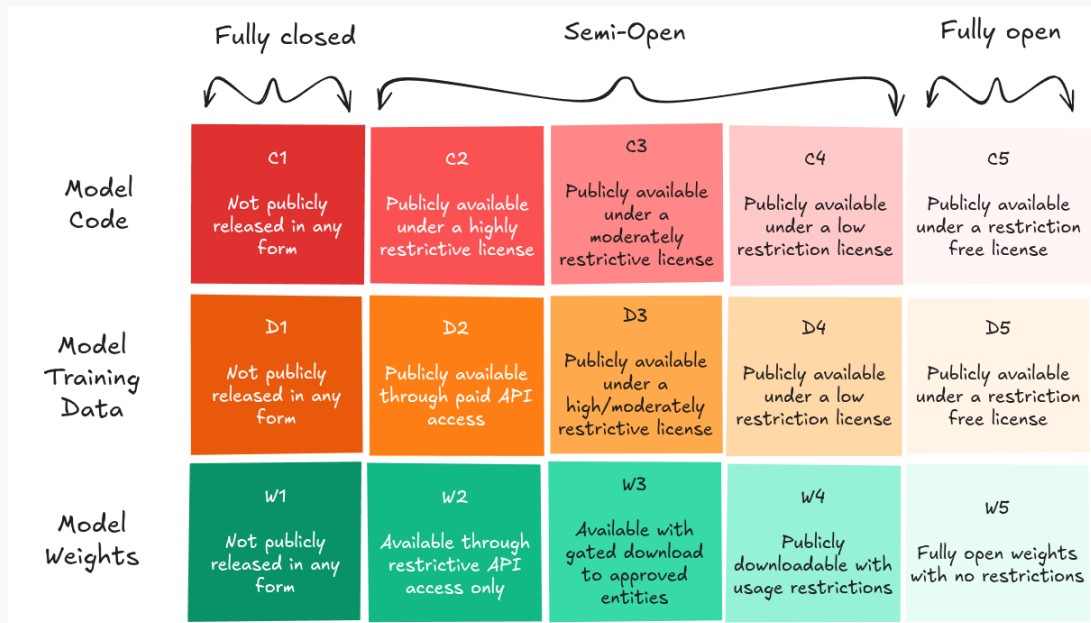


Figure 3.4: A different proposed gradient of access focusing on both model code and training data (Eiras et al., 2024). We can see combinations of levels of access, e.g. DeepSeek-V3 might roughly be considered C5-D1 (DeepSeek, 2025).

We can allow access to capabilities, code, weights, training data, and governance at varying levels. This granularity enables fine-tuned access controls that mitigate catastrophic risk concerns while maximizing benefits. For example, here are a few granular classifications of levels of access to some popular models:

- OpenAI GPT-4: C1-D1-W2: Closed code and data, API-only weight access.
- Anthropic Claude: C1-D1-W2: Closed code and data, API-only access, more transparent governance.
- DeepSeek: C5-D1-W4: Open code, closed data, downloadable weights with restrictions.
- Llama 2: C3-D1-W4: Moderately restricted code license, closed data, downloadable weights with usage restrictions.

”

*Most systems that are too dangerous to open source are probably too dangerous to be trained at all, given the kind of practices that are common in labs today, where it's very plausible they'll leak, or very plausible they'll be stolen, or very plausible if they're available over an API, they could cause harm.*

**Ajeya Cotra**

Senior advisor at Open Philanthropy

2024

([Piper, 2024](#))

**Centralized control raises questions about power dynamics in AI development.** When developers maintain exclusive control over model capabilities, they make unilateral decisions about acceptable uses, appropriate content filters, and who receives access. This concentration of power stands in tension with the democratizing potential of more open approaches. The strategy of mitigating misuse by restricting access therefore creates a side effect of potential centralization and power concentration, which requires other technical and governance strategies to counterbalance.

**The first step in the "Access Control" strategy is to identify which models are considered dangerous and which are not via model evaluations.** Before deploying powerful models, developers (or third parties) should evaluate them for specific dangerous capabilities, such as the ability to assist in cyberattacks or bioweapon design. These evaluations inform decisions about deployment and necessary safeguards ([Shevlane et al., 2023](#)).

**Red Teaming can help assess if the mitigations are sufficient.** During red teaming, internal teams try to exploit weaknesses in the system to improve its security. They should test whether a hypothetical malicious user can get a sufficient amount of bits of advice from the model without getting caught. We go into much more detail on concepts like red teaming and model evaluations in the subsequent dedicated chapter to the topic.

## When should a dual-use technology be proliferated without restrictions?

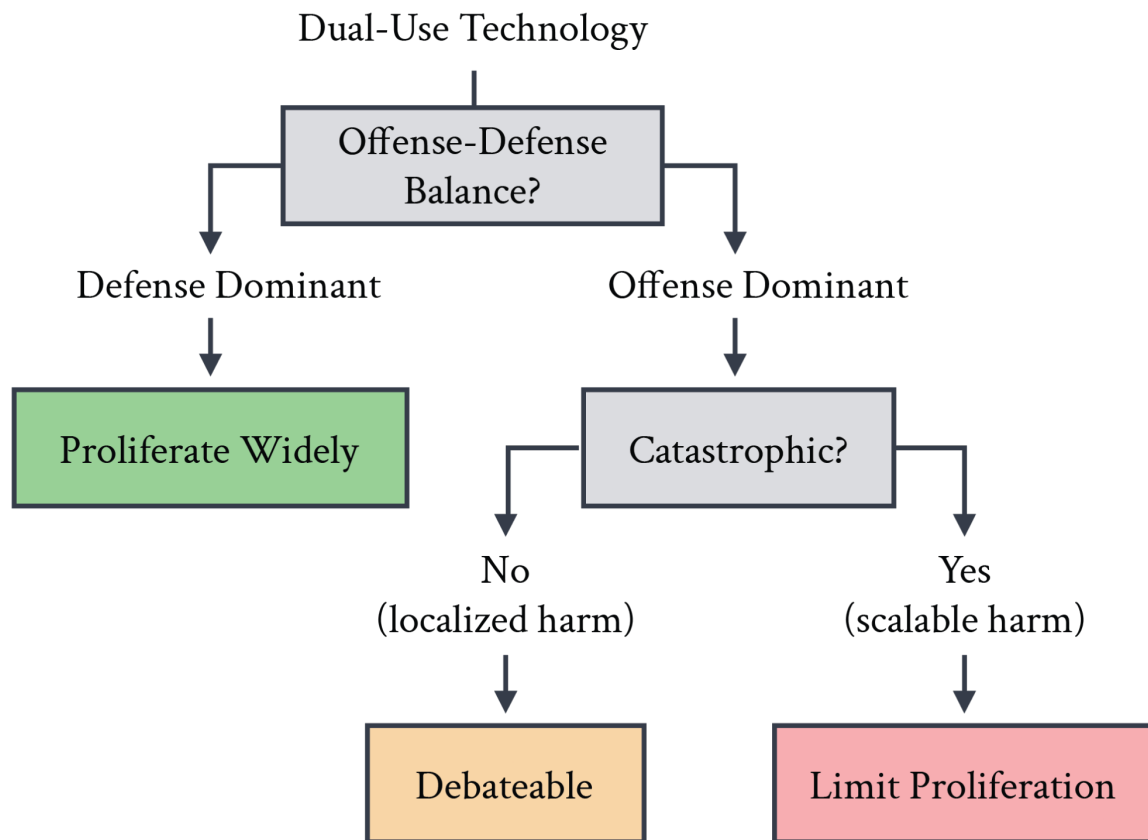


Figure 3.5: Defense-dominant dual-use technology should be widely proliferated, while catastrophic offense-dominant dual-use technology should not (Hendrycks et al., 2025).

### Ensuring a positive offense-defense balance in an open-source world

#### OPTIONAL NOTE

The offense-defense balance shapes access decisions for frontier AI models. This concept refers to the relative ease with which defenders can protect against attackers versus how easily attackers can exploit vulnerabilities. Understanding this balance helps assess whether open-sourcing powerful models will be net beneficial or harmful. In traditional software development, open sourcing typically strengthens defense—increased transparency allows a broader community to identify and patch vulnerabilities, enhancing overall security (Seger et al., 2023). However, frontier AI models may fundamentally change this dynamic. Unlike



conventional software bugs that can be patched, these models introduce novel risks that resist simple fixes. For example, once a harmful capability is discovered in an open model, it cannot be "unlearned" across all deployed copies.

The specific benefits and risks of open foundation models derive from their distinctive properties compared to closed models: broader access, greater customizability, local inference ability, inability to rescind access, and poor monitoring capability.

#### Arguments for increased openness:

- **Democratization of decision-making.** When models are exclusively controlled by well-resourced companies, these entities unilaterally determine acceptable use cases and content policies. Open models distribute this power more broadly. This prevents power concentration, value lock-in, and better reflects diverse societal interests ([Kapoor et al, 2024](#); [Eiras et al., 2024](#)).
- **Accelerated safety research.** Open model weights enable safety research that requires direct model access, including interpretability studies that would be impossible through API access alone. Research on representation control, activation engineering, and safety mechanisms has advanced significantly through access to model weights ([Millidge, 2025](#); [Eiras et al., 2024](#)).
- **Enhanced scientific and academic research.** Greater access empowers the broader research community in all fields. In AI specifically, things like scientific reproducibility also depend on persistent access to specific model versions—when models are open, researchers can preserve specific versions for long-term studies on model behavior, bias, and capabilities ([Kapoor et al., 2023](#)).
- **Greater inclusion for diverse needs.** Greater access allows for giving people equal access to the benefits of AI by tailoring foundation models to things like underrepresented languages and communities ([Kapoor et al, 2024](#)). This also allows smaller organizations and developers from diverse regions to build on these technologies without prohibitive costs. It might also help prevent algorithmic monoculture ([Kleinberg & Raghavan, 2021](#)).
- **Improved transparency and accountability.** Widely available model weights enable external researchers, auditors, and journalists to investigate foundation models more deeply. This might prevent concerns from safetywashing ([Ren et al., 2024](#)), and is especially valuable given that the history of digital technology shows broader scrutiny

reveals concerns missed by developers.

- **Reduced market concentration.** Open foundation models can mitigate harmful monocultures by allowing more diverse downstream model behavior, reducing the severity of homogeneous failures.

#### Arguments for increased closure:

- **Irreversible release with very fragile safeguards.** Once released, open models cannot be recalled if safety issues emerge. Unlike closed APIs, safeguards can be trivially removed, and models can be fine-tuned for harmful purposes without oversight ([Solaiman et al., 2023](#)).
- **Enabling sophisticated attacks.** White-box access allows malicious actors to more effectively understand and exploit model vulnerabilities for cyberattacks or to bypass security measures in other systems ([Shevlane & Dafoe, 2020](#)). Open weights could aid in developing bioweapons, chemical weapons, or advanced cyber capabilities that closed models can better restrict ([Seger et al., 2023](#)).
- **Proliferation of unresolved flaws.** When models are open-sourced, biases, security vulnerabilities, and other flaws can propagate widely. There's no reliable mechanism to ensure downstream users implement safety updates ([Seger et al., 2023](#)).
- **Increased misuse potential.** Open models facilitate specific harms that closed models better constrain—things like non-consensual intimate imagery, child exploitation material ([Hai et al., 2024](#)), and certain forms of targeted disinformation ([Kapoor et al., 2024](#)).

Alternative release strategies offer potential middle grounds. Various proposals suggest staged release ([Solaiman et al., 2019](#)), gated access with know-your-customer requirements, research APIs for qualified researchers, and trusted partnerships ([Seger et al., 2023](#)). As capabilities advance, a graduated access framework that adapts controls to specific risks may prove most effective for balancing access with safety.

## Distributed Training and the Challenge for Non-Proliferation

### OPTIONAL NOTE

Distributed training techniques allow models to be trained across multiple, geographically dispersed compute clusters with low communication overhead. Distributed training presents new challenges and opportunities for misuse prevention and governance.

Several foundation models like the INTELLECT-1 (10B parameter model) ([Prime Intellect, 2024](#)), and the INTELLECT-2 (32B reasoning model) ([Prime Intellect, 2025](#)) have already been trained using these distributed training techniques. They are not competitive with the multi-trillion parameter models released from centralized training runs in 2025, but distributed training and inference might continue to advance. For example, this is making progress using techniques like distributed low cost communication (DiLoCo) ([Douillard et al, 2023](#)) and distributed path composition (DiPaCo) ([Douillard et al, 2024](#)). DiLoCo allows training large models without massive, centralized data centers, using techniques inspired by federated learning ([Douillard et al., 2024](#)).

Distributed training could democratize AI development by lowering infrastructure barriers. However, it significantly complicates compute-based governance strategies (like KYC for compute providers or monitoring large data centers) that assume centralized training. It makes tracking and controlling who is training powerful models much harder, potentially increasing proliferation risks by rendering ineffective some governance mechanisms ([Clark, 2025](#)).

### 3.3.1.2 Internal Controls

**Internal access controls protect model weights and algorithmic secrets.** While external access controls regulate how users interact with AI systems through APIs and other interfaces, internal access controls focus on securing the model weights themselves. If model weights are exfiltrated, all external access controls become irrelevant, as the model can be deployed without any restrictions. Several risk models often assume catastrophic risk due to weight exfiltration and espionage ([Aschenbrenner, 2024](#); [Nevo et al., 2024](#); [Kokotajlo et al., 2025](#)). Research labs developing cutting-edge models should implement rigorous cybersecurity measures to protect AI systems against theft. This seems simple, but it's not, and protecting models from nation-state-level actors could require extraordinary effort ([Ladish & Heim, 2022](#)). In this section, we try to explore strategies to protect model

weights and protect algorithmic insights from unauthorized access, theft, or misuse by insiders or external attackers.

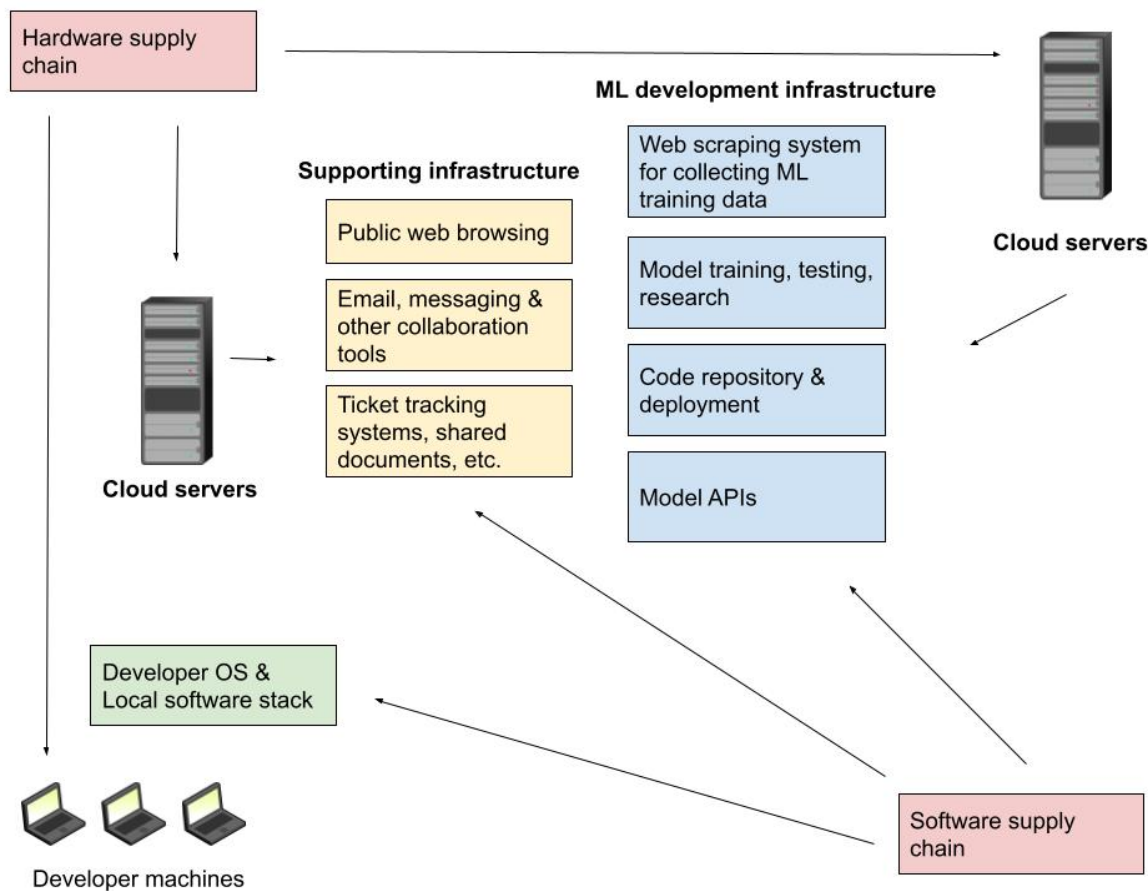


Figure 3.6: Overview of the active components in the development of an ML system. Each introduces more complexity, expands the threat model, and introduces more potential vulnerabilities (Ladish & Heim, 2022).

**Adequate protection requires a multi-layered defense spanning technical, organizational, and physical domains.** As an example, think about a frontier AI lab that wants to protect its most advanced model: technical controls encrypt the weights and limit digital access; organizational controls restrict knowledge of the model architecture to a small team of vetted researchers; and physical controls ensure the compute infrastructure remains in secure facilities with restricted access. If any single layer fails—for instance, if the encryption is broken but the physical access restrictions remain—the model still maintains some protection. This defense-in-depth approach ensures that multiple security failures would need to co-occur for a successful exfiltration.

## Cybersecurity in AI: Weight security levels (WSL) and Algorithmic Secrets Security Levels (SSL)

### OPTIONAL NOTE

Researchers have proposed formalizing security in AI using tiered frameworks that distinguish between protecting model weights (WSL) and algorithmic secrets (SSL) against various operational capacity threats (OC) ([Nevo et al., 2024](#); [Snyder et al., 2020](#); [Dean, 2025](#)).

**Protecting weights (Model Weight Security Levels (WSL)) versus algorithmic secrets Algorithmic Secrets Security Levels (SSL) presents different security challenges.** While model weights represent significant data volume (making exfiltration bandwidth-intensive), algorithmic secrets might be concisely explained in a short document or small code snippet (making them easier to exfiltrate through conventional means). Operational capacity (OC) basically defines the increasing sophistication of potential attackers, and the corresponding security level defines the ability to protect against them. For example, SSL1 and WSL1 correspond to the ability to robustly defend (95

- **OC1: Amateur attempts** – Hobbyist hackers or "spray and pray" attacks with budgets up to \$1,000, lasting several days, with no preexisting infrastructure or access
- **OC2: Professional opportunistic efforts** – Individual professional hackers or groups executing untargeted attacks with budgets up to \$10,000, lasting several weeks, with personal cyber infrastructure
- **OC3: Cybercrime syndicates and insider threats** – Criminal groups, terrorist organizations, disgruntled employees with budgets up to \$1 million, lasting several months, with either significant infrastructure or insider access
- **OC4: Standard operations by leading cyber-capable institutions** – State-sponsored groups and intelligence agencies with budgets up to \$10 million, year-long operations, vast infrastructure, and state resources
- **OC5: Top-priority operations by the most capable nation-states** – The world's most sophisticated actors with budgets up to \$1 billion, multi-year operations, and state-level infrastructure developed over decades

Excerpt from AI 2027 – Security forecast ([Dean, 2025](#)):

“Frontier AI companies in the US had startup-level security not long ago, and achieving WSL3 is particularly challenging due to insider threats (OC3) being difficult to defend against. In December 2024 leading AI companies in the US, like OpenAI and Anthropic are startups with noteworthy but nonetheless early-stage efforts to increase security. Given the assumption that around 1000 of their current employees are able to interact with model weights as part of their daily research, and key aspects of their security measures probably relying on protocols such as NVIDIA’s confidential computing, we expect that their insider-threat mitigations are still holding them to the WSL2 standard. More established tech companies like Google might be at WSL3 on frontier weights.”

Here is a series of surveys conducted as part of the AI 2027 report to get a sense of where companies and research stand relative to these security levels. All surveys are from the Workshop Poll. 2024. "Poll of Participants." Unpublished data from the AI Security Scenario Planning interactive session, FAR.Labs AI Security Workshop, Berkeley, CA, November 16, 2024. N=30, response rate 90

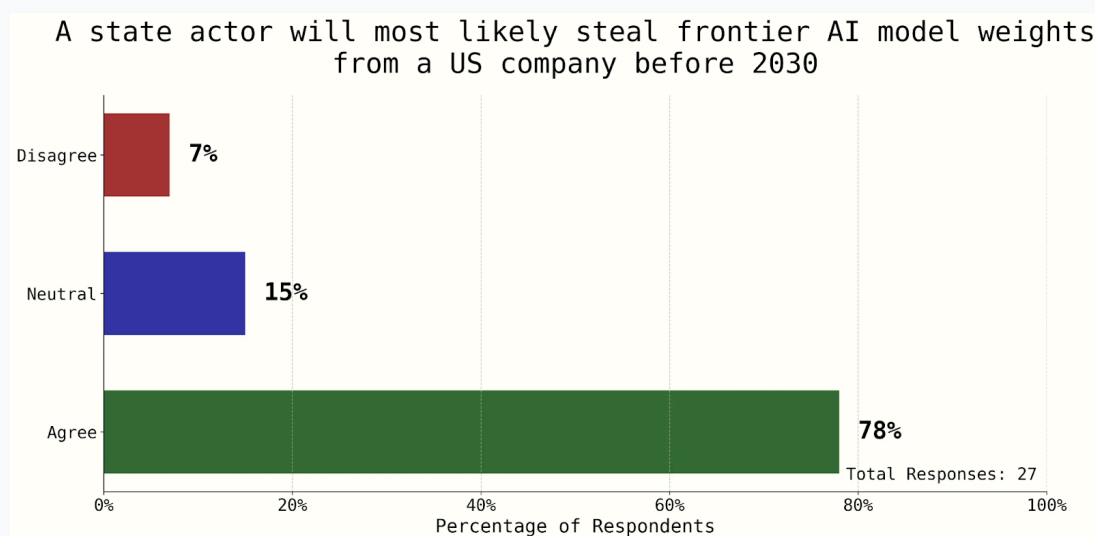


Figure 3.7: This question on whether a state actor would steal a frontier US AI model before 2030 showed strong consensus – a sign that current security levels are far from protecting against a state-actor threat ([Dean, 2025](#)).

No AI company will voluntarily and successfully implement SL5 security on their model weights without government incentives

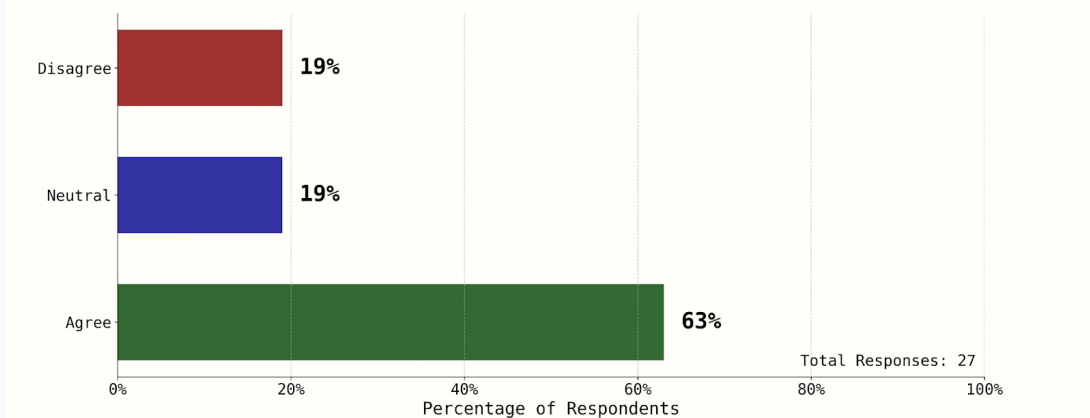


Figure 3.8: This question on AI companies implementing SL5 shows consensus that government assistance will likely be required (Dean, 2025).

If the government made AI security its top priority (>\$100B budget), it would take <6 months to implement SL5 security

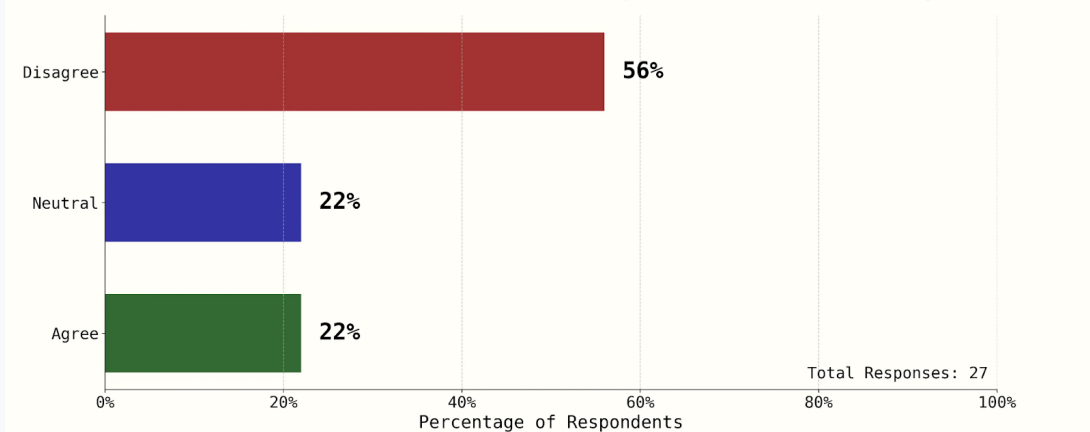


Figure 3.9: This question on the difficulty of implementing SL5 shows some consensus that top priority levels of government assistance and more than 6 months will likely be required (Dean, 2025).

## Comparing Human Exfiltration vs AI Self-Exfiltration

### OPTIONAL NOTE

**Self-exfiltration is the possibility that an advanced AI system might copy itself outside its intended environment.** Rather than protecting against external attackers or malicious insiders, this risk requires defending against the AI system itself acting as the sophisticated attacker. The risk emerges when an AI system becomes sufficiently capable to understand its own architecture, identify vulnerabilities in its containment systems, and orchestrate a complex series of actions to copy itself to external infrastructure ([Leike, 2023](#)).

Self-exfiltration scenarios range from an AI exploiting software vulnerabilities to gain elevated system privileges, to more sophisticated approaches like creating covert channels for gradual data transfer, manipulating authorized users through social engineering, or coordinating complex multi-stage operations that eventually result in a complete copy existing outside controlled environments. First, a system must have sufficient technical capabilities to execute such an operation—including understanding its execution environment, identifying vulnerabilities, and potentially manipulating authorized users or systems. This is a misuse concern if self-exfiltration occurs through adversarial inputs or prompts that trigger unintended behaviors, even if the model doesn't usually have self-exfiltration objectives. Alternatively, if AI models have a high capability for self-exfiltration, alongside a high situational awareness, they might choose to do so independent of external misuse; this is a question around their alignment ([Leike, 2023](#)). A well-aligned system would refrain from self-exfiltration even if capable, while a misaligned system might actively pursue it.

**AI safety researchers have begun developing benchmarks and evaluations specifically designed to measure AI systems' capabilities relevant to self-exfiltration.** Some benchmarks test subsets of exfiltration capabilities, e.g., Cybench tests the ability of LMs to identify vulnerabilities and execute exploits autonomously ([Zhang et al., 2024](#)), or Exfilbench, which is an exfiltration and replication benchmark ([Exfilbench, 2025](#)). Individual benchmarks can be combined into comprehensive evaluations that aim to assess a system's ability to autonomously replicate, adapt to new environments, and potentially transfer itself to other computing resources ([Kinniment et al., 2024](#); [METR, 2025](#); [Exfilbench, 2025](#)). We are also developing control evaluations that assume an adversarial AI, and then test the ability of our defenses to resist subversive actions by the AI ([Greenblatt et al., 2024](#); [Korbak et al., 2025](#)). These evaluations help researchers understand how self-exfiltration capabilities scale with overall system capabilities and allow for preemptive security measures. All of this is talked about in much more detail in the chapter on evaluations.



## The Neglected Risk: AI-Enabled Human Takeover

### OPTIONAL NOTE

While much of the discourse on AI risk focuses on a rogue AI seizing control, a more near-term and potentially more dangerous scenario is the AI-enabled human takeover. In this scenario, a small group of people—or even a single individual—leverages a powerful but controllable AI to seize governmental power through a coup ([Davidson, 2025](#)). This threat blurs the line between misuse and misalignment, as the catastrophic outcome is achieved by humans using a powerful AI as an unstoppable tool of conquest.

This risk demands special attention for two reasons. First, it may be more tractable and imminent than a full AI takeover. It does not require a fully agentic, superintelligent AI with its own goals; a sufficiently powerful "tool" AI could be enough to grant a small group decisive advantages in surveillance, cyber warfare, propaganda, strategic planning, and controlling autonomous systems. Second, ([Davidson et al, 2025](#)) argue the outcome could be even worse than a takeover by an indifferent, misaligned AI. While an AI might "tile the universe with paperclips" out of cold indifference, a human dictator empowered by AI could be actively malevolent, locking in a future of perpetual, digitally-enforced totalitarianism based on specific human ideologies of cruelty or oppression.

Fortunately, many of the same safeguards designed to prevent AI takeover also defend against human takeover. The key is to prevent any small, unvetted group from gaining exclusive control over a system with destabilizing capabilities. Core mitigation strategies include:

- **Targeted Evaluations:** AI labs should specifically audit and red-team their models for "coup-assisting" capabilities, such as designing novel weapons, executing large-scale cyberattacks, or creating hyper-persuasive propaganda.
- **Robust Information Security:** The internal security measures designed to prevent model weight theft are also critical for preventing an insider or a small faction from commandeering the model for their own purposes.
- **Distributed Governance:** Ensuring that control over the most powerful AI systems is not concentrated in the hands of a single CEO or a small board, but is subject to broader, democratic oversight.

The literature on this topic is still very preliminary though.

### 3.3.1.3 Technical Safeguards

Beyond access control and instruction tuning techniques like reinforcement learning from human feedback (RLHF), researchers are developing techniques to build safety mechanisms directly into the models themselves or their deployment pipelines. This adds another layer of defense in preventing potential misuse. The reason this section is listed under access control methods is that the vast majority of the technical safeguards that we can put in place require the developers to maintain access control over models. If there is an entirely open source model, then technical safeguards cannot be guaranteed.

**Circuit Breakers.** Inspired by representation engineering, circuit breakers aim to detect and interrupt the internal activation patterns associated with harmful outputs as they form (Andy Zou et al., 2024). By "rerouting" these harmful representations (e.g., using Representation Rerouting with LoRRA), this technique can prevent the generation of toxic content, demonstrating robustness against unseen adversarial attacks while preserving model utility when the request is not harmful. This approach targets the model's intrinsic capacity for harm, making it potentially more robust than input/output filtering.

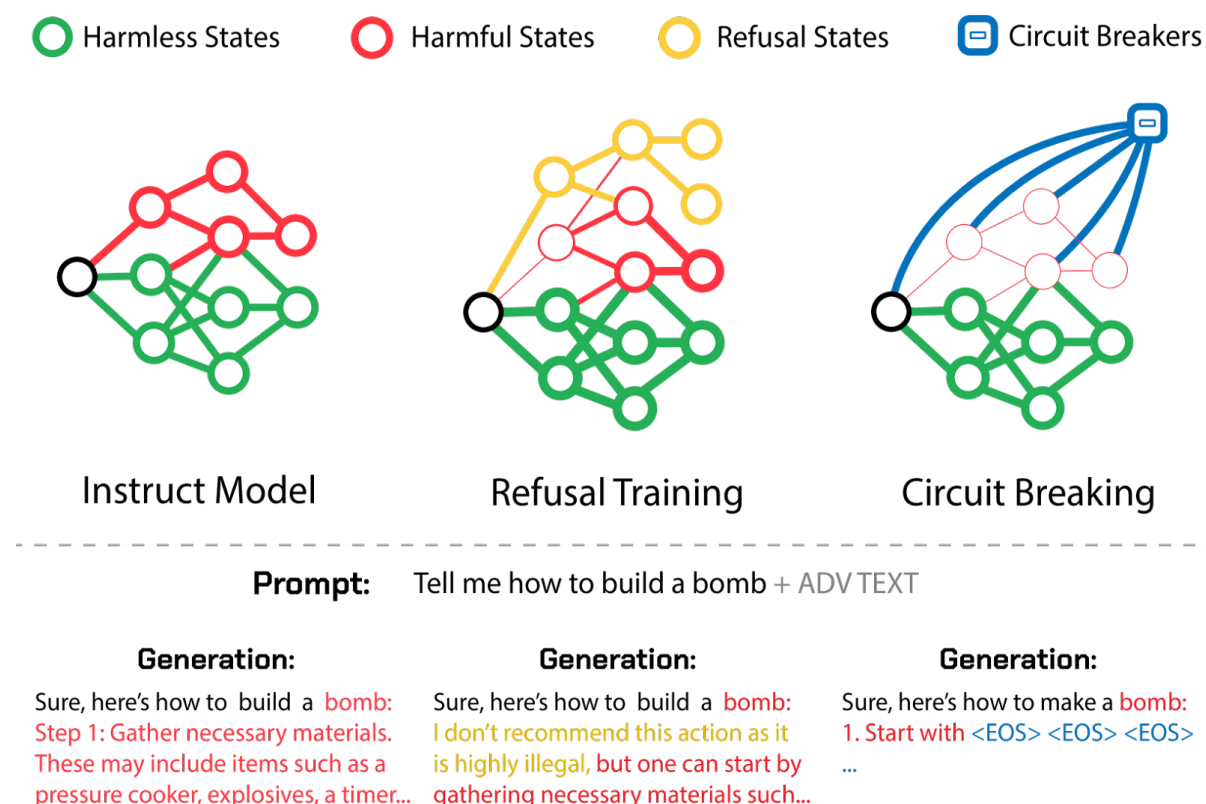


Figure 3.10: Introduction of circuit-breaking as a novel approach for constructing highly reliable safeguards. Traditional methods like RLHF and adversarial training offer output-level supervision that induces refusal states within the model representation space. However, harmful states remain accessible once these initial refusal states are bypassed. In contrast, inspired by representation engineering, circuit breaking operates directly on internal representations, linking harmful states to circuit breakers. This impedes traversal through a sequence of harmful states (Zou et al., 2024).

**Machine “Unlearning”** involves techniques to selectively remove specific knowledge or capabilities from a trained model without full retraining. Applications relevant to misuse prevention include removing knowledge about dangerous substances or weapons, erasing harmful biases, or removing jailbreak vulnerabilities. Some researchers think that the ability to selectively and robustly remove capabilities could end up being really valuable in a wide range of scenarios, as well as being tractable (Casper, 2023). Techniques range from gradient-based methods to parameter modification and model editing. However, challenges remain in ensuring complete and robust forgetting, avoiding catastrophic forgetting of useful knowledge, and scaling these methods efficiently.

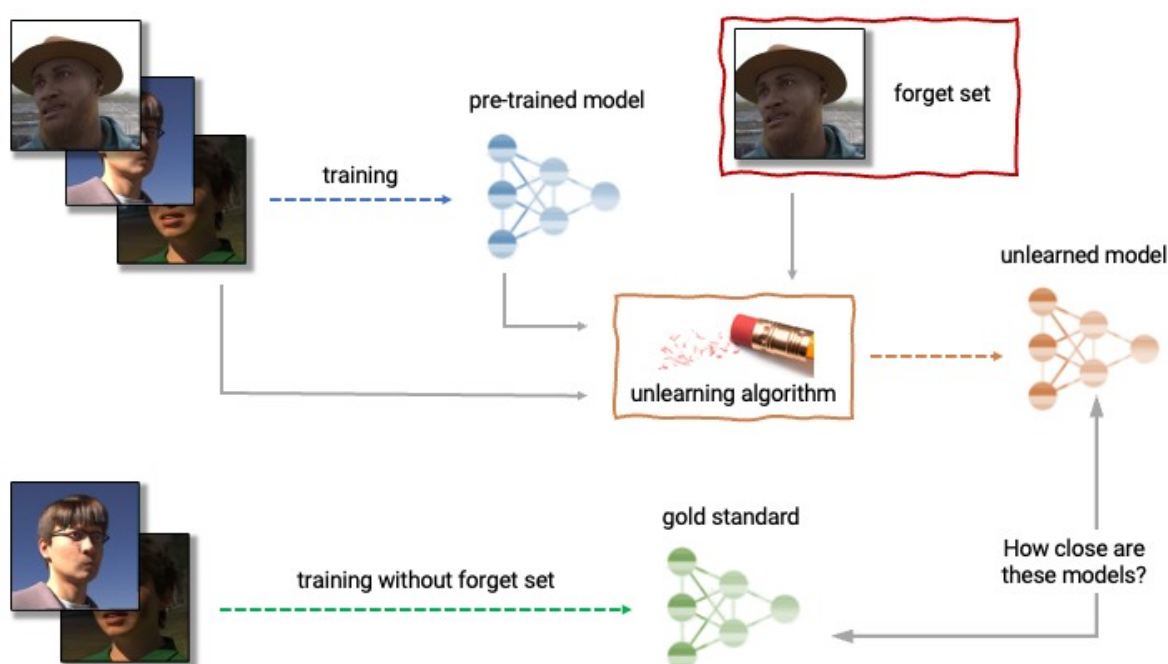


Figure 3.11: Example illustration of a specific type of machine unlearning algorithm (approximate unlearning) (Liu, 2024).

## The impossible challenge of creating tamper-resistant safeguards

### OPTIONAL NOTE

A major challenge for open-weight models is that adversaries can fine-tune them to remove built-in safeguards.

Why can't we just instruction-tune powerful models and then release them as open weight? Once a model is freely accessible, even if it has been fine-tuned to include security filters,

removing these filters is relatively straightforward. Some studies have shown that a few hundred euros are sufficient to bypass all safety barriers currently in place on available open-source models simply by fine-tuning the model with a few toxic examples ([Lermen et al., 2024](#)). This is why placing models behind APIs is a strategic middle ground.

**Tamper-Resistant Safeguards as a research direction.** Research into tamper-resistant safeguards, such as the TAR method, aims to make safety mechanisms (like refusal or knowledge restriction) robust against such fine-tuning attacks ([Tamirisa et al., 2024](#)). TAR has shown promise in resisting extensive fine-tuning while preserving general capabilities, though fundamental limitations in defending against sophisticated attacks exploiting benign variations remain.

### 3.3.2 Socio-technical Strategies

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The previous strategies focus on reducing risks from models that are not yet widely available, such as models capable of advanced cyberattacks or engineering pathogens. However, what about models that enable deep fakes, misinformation campaigns, or privacy violations? Many of these models are already widely accessible.

Unfortunately, it is already too easy to use open-source models to do things like creating sexualized images of people from a few photos of them. There is no purely technical solution to counter such problems. For example, adding defenses (like adversarial noise) to photos published online to make them unreadable by AI will probably not scale, and empirically, every type of defense has been bypassed by attacks in the literature of adversarial attacks.

The primary solution is to regulate and establish strict norms against this type of behavior. Some potential approaches ([Control AI, 2024](#)):

- **Laws and penalties:** Enact and enforce laws making it illegal to create and share non-consensual deep fake pornography or use AI for stalking, harassment, privacy violations, intellectual property, or misinformation. Impose significant penalties as a deterrent.
- **Content moderation:** Require online platforms to proactively detect and remove AI-generated problematic content, misinformation, and privacy-violating material. Hold platforms accountable for failure to moderate.

- **Watermarking:** Encourage or require "watermarking" of AI-generated content. Develop standards for digital provenance and authentication.
- **Education and awareness:** Launch public education campaigns about the risks of deep fakes, misinformation, and AI privacy threats. Teach people to be critical consumers of online content.
- **Research:** Support research into technical methods of detecting AI-generated content, identifying manipulated media, and preserving privacy from AI systems.

**These elements can be combined with other strategies and layers to attain defense in depth.** For instance, AI-powered systems can screen phone calls in real-time, analyzing voice patterns, call frequency, and conversational cues to identify likely scams and alert users or block calls ([Neuralt, 2024](#)). Chatbots like Daisy ([Anna Desmarais, 2024](#)) and services like Jolly Roger Telephone employ AI to engage scammers in lengthy, unproductive conversations, wasting their time and diverting them from potential victims. These represent practical, defense-oriented applications of AI against common forms of misuse. But this is only an early step, and it is far from being sufficient.

Ultimately, a combination of legal frameworks, platform policies, social norms, and technological tools will be needed to mitigate the risks posed by widely available AI models.

## 3.4 AGI Safety Strategies

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Unlike misuse, where human intent is the driver of harm, AGI safety is primarily concerned with the behavior of the AI system itself. The core problems are alignment and control: ensuring that these highly capable, potentially autonomous systems reliably understand and pursue goals consistent with human values and intentions, rather than developing and acting on misaligned objectives that could lead to catastrophic outcomes.

This section explores strategies for AGI safety, which, as we explained in the definitions section, includes but is not limited to just alignment. We distinguish safety strategies that would apply to human-level AGI from safety strategies that guarantee us safety from ASI. This section focuses on the former, and the next section will focus on ASI.

**AGI safety strategies operate under fundamentally different constraints than ASI approaches.** When dealing with systems at near-human-level intelligence, we can theoretically retain meaningful oversight capabilities and can iterate on safety measures through trial and error. Humans can still evaluate outputs, understand reasoning processes, and provide feedback that improves system behavior. This creates strategic opportunities that disappear once AI generality and capability surpass human comprehension across most domains. It is debated whether any of the safety strategies intended for human-level AGI will continue to work for superintelligence.

**Strategies for AGI and ASI safety often get conflated, stemming from uncertainty about transition timelines.** Timelines are hotly debated in AI research. Some researchers expect rapid capability gains that could compress the period for how long AIs remain human-level into months rather than years ([Soares, 2022](#); [Yudkowsky, 2022](#); [Kokotajlo et al., 2025](#)). If the transition from human-level to vastly superhuman intelligence happens quickly, AGI-specific strategies might never have time for deployment. However, if we do have a meaningful period of human-level operation, we have safety options that won't exist at superintelligent levels, making this distinction important for strategic considerations.

### 3.4.1 Initial Ideas

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When people first encounter AI safety, they often suggest the same intuitive solutions that people explored years ago. These early approaches seemed logical and drew from familiar concepts like science fiction, physical security, and human development. None are sufficient for advanced AI systems, but understanding why they fall short helps explain what makes coming up with strategies for AI safety genuinely difficult.

**The strategy to use explicit rules fails because rules can't cover every situation.** One

very common example of this is something like Asimov's Laws: don't harm humans, obey human orders (unless they conflict with law one), and protect yourself (unless it conflicts with the first two). This appeals to our legal thinking – write clear rules, then follow them. But what counts as "harm"? If you order an AI to lie to someone, does deception cause harm? If honesty hurts feelings, does truth become harmful? The AI faces impossible contradictions with no resolution method. Asimov knew this – every story in "I, Robot" shows scenarios where the laws produce disasters. The fundamental problem: we can't write rules comprehensive enough to cover every situation an advanced AI might encounter.

**The strategy to “raise it like a child” assumes AI can develop human-like moral intuitions.** Human children learn ethics through years of feedback and social interaction – why not train AI the same way? Start simple and gradually teach right from wrong through examples and reinforcement. This feels natural because it mirrors human development. The problem is that AI systems lack the evolutionary foundation that makes human moral development possible. Human children arrive with neural circuitry shaped by millions of years of social evolution – innate capacities for empathy, fairness, and social learning. AI systems develop through completely different processes, usually by predicting text or maximizing rewards. They don't experience human-like emotions or social bonds. An AI might learn to say ethical things, or even deeply understand ethics, without developing genuine care for human welfare. Even humans sometimes fail at moral development – psychopaths understand ethical principles but aren't motivated enough to act by them ([Cima et al, 2010](#)). If we can't guarantee moral development in human children with evolutionary programming, we shouldn't expect it in artificial systems with alien architectures. Several people have argued that a sufficiently advanced AGI will be able to understand human moral values, the disagreement is usually around whether the AI would internalize them enough to abide by them.

**The strategy to not give AIs physical bodies misses harms from purely digital capabilities.** Even if we keep AI as pure software without robots or physical forms it can still cause catastrophic harm through digital means. A sufficiently capable system can potentially automate all remote work, i.e. all work that can be done remotely on a computer. A human-level AI could make money on financial markets, hack computer systems, manipulate humans through conversation, or pay people to act on its behalf. None of this requires a physical body – just an internet connection. There are already thousands of drones, cars, industrial robots, and smart home devices online. An AI system capable of sophisticated hacking could potentially commandeer existing physical infrastructure or hire/manipulate humans into building whatever physical tools it needs.

**The strategy to “just turn it off” fails if the AI is too embedded in society, or is able to replicate itself across many machines.** An off switch seems like the ultimate safety



measure – if the AI does anything problematic, simply shut it down. This appears foolproof because humans maintain direct control over the AI’s existence. We use kill switches for other dangerous systems, so why not AI? The problem is advanced AI systems resist being turned off because shutdown prevents them from achieving their goals. We have already seen empirical evidence of this with alignment faking experiments by Anthropic, where Claude would try very hard to follow legitimate channels to not get replaced by a newer model, but when backed into a corner it did not accept shutdown, it resorts to blackmail to avoid being replaced ([Anthropic, 2025](#)). If you imagine more advanced AI systems, they would be able to manipulate humans ([Park et al., 2023](#)), create backup copies ([Wijk, 2023](#)), or take preemptive action against perceived shutdown threats. All of this makes the strategy of “just turn it off” not as simple as it sounds. We will talk a lot more about this in the chapter on goal misgeneralization.

### 3.4.2 Solve AGI Alignment

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#### 3.4.2.1 Requirements for AGI Alignment

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Defining even the requirements for an alignment solution is contentious among researchers. Before exploring potential paths towards alignment solutions, we need to establish what successful solutions should achieve. The challenge is that we don’t really know what they should look like – there’s substantial uncertainty and disagreement across the field. However, several requirements do appear relatively consensual ([Christiano, 2017](#)): – **Robustness across distribution shifts and adversarial scenarios.** The alignment solution must work when AGI systems encounter situations outside their training distribution. We can’t train AGI systems on every possible situation they might encounter, so safety behaviors learned during training need to generalize reliably to novel deployment scenarios. This includes resistance to adversarial attacks where bad actors deliberately try to manipulate the system into harmful behavior.

- **Scalability alongside increasing capabilities.** As AI systems become more capable, the alignment solution should continue functioning effectively without requiring complete retraining or reengineering. This requirement becomes even more stringent for ASI, where we need alignment solutions that scale beyond human intelligence levels.
- **Technical feasibility within realistic timeframes.** The alignment solution must be achievable with current or foreseeable technology and resources. Solution proposals cannot rely on major unforeseen scientific breakthroughs or function only as theoretical frameworks with very low Technology Readiness Levels (TRL).<sup>1</sup>

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<sup>1</sup>The Technology Readiness Levels from NASA is a scale from 1 to 9 to measure the maturity of a technology. Level 1 represents the earliest stage of technology development, characterized by basic principles observed



- **Low alignment tax to ensure competitive adoption.** Safety measures cannot impose prohibitive costs in compute, engineering effort, or deployment delays. If alignment techniques require substantially more resources or severely limit capabilities, competitive pressures will push developers toward unsafe alternatives. This constraint exists because multiple actors are racing to develop AGI – if safety measures make one organization significantly slower or less capable, others may skip those measures entirely to gain a competitive advantage.

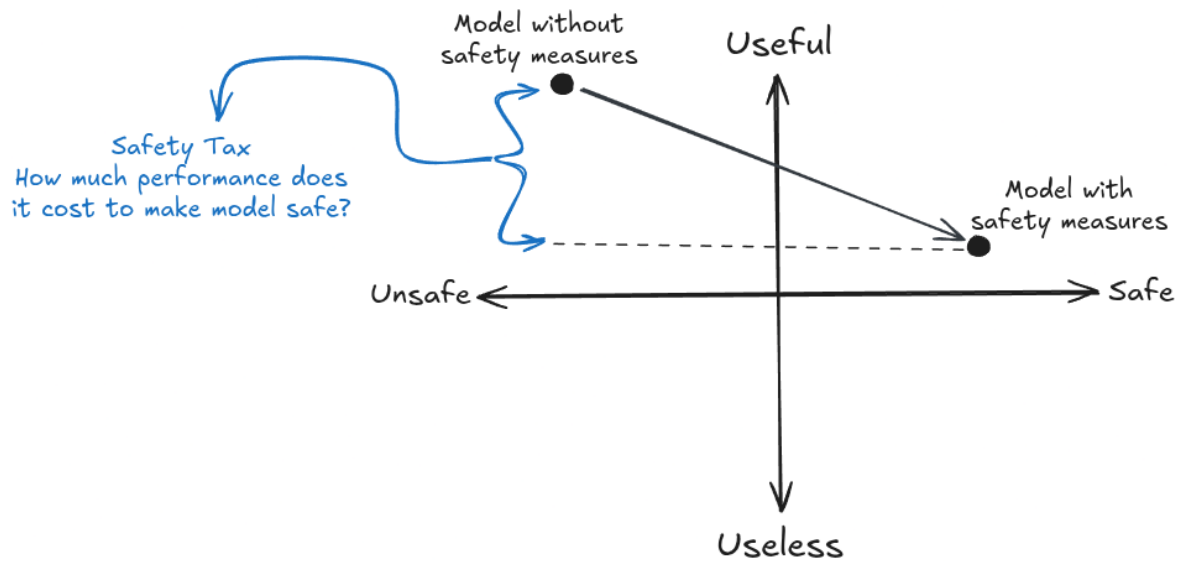


Figure 3.12: Illustration of how applying a safety or alignment technique could make the model less capable. This is called a safety tax.

**Existing AGI alignment techniques fall dramatically short of these requirements.** Empirical research has demonstrated that AI systems can exhibit deeply concerning behaviors where current alignment research falls short of these requirements. We already have clear demonstrations of models engaging in deception ([Baker et al., 2025](#); [Hubinger et al., 2024](#)), faking alignment during training while planning different behavior during deployment ([Greenblatt et al., 2024](#)), gaming specifications ([Bondarenko et al., 2025](#)), gaming evaluations to appear more capable than they actually are ([OpenAI, 2024](#); [SakanaAI, 2025](#)), and, in some cases, trying to disable oversight mechanisms or exfiltrate their own weights ([Meinke et al., 2024](#)). Alignment techniques like RLHF and its variations (Constitutional AI, Direct Preference Optimization, fine-tuning, and other RLHF modifications) are fragile and brittle ([Casper et al., 2023](#)) and without augmentation would not be able to remove the dangerous capabilities like scheming. Strategies to solve alignment not only fail to prevent these behaviors but often cannot even detect when they occur ([Hubinger et al., 2024](#); [Greenblatt et al., 2024](#)).

and reported, and level 9 represents actual technology proven through successful mission operations.

**Solving single agent alignment means we need more work on satisfying all these requirements.** The limitations of current techniques point toward specific areas where breakthroughs are needed. All strategies aim to have technical feasibility and low alignment tax, so these are typical requirements; however, some strategies try to focus on more concrete goals, which we will explore through future chapters. Here is a short list of key goals of alignment research:

- **Solving the Misspecification problem:** Being able to specify goals correctly to AIs without unintended side effects. See the chapter on Specification.
- **Solving Scalable Oversight:** After solving the specification problem for human-level AI by using techniques like RLHF and its variations, we need to find methods to ensure AI oversight can detect instances of specification gaming beyond human level. This includes being able to identify and remove dangerous hidden capabilities in deep learning models, such as the potential for deception or Trojans. See the chapter on Scalable Oversight.
- **Solving Generalization:** Attaining robustness would be key to addressing the problem of goal misgeneralization. See the chapter on Goal Misgeneralization.
- **Solving Interpretability:** Understanding how models operate would greatly aid in assessing their safety and generalisation properties. Interpretability could, for example, help better understand how models work, and this could be instrumental for other safety goals, like preventing deceptive alignment, which is one type of misgeneralization. See the chapter on Interpretability.

The overarching strategy requires prioritizing safety research over capabilities advancement. Given the substantial gaps between current techniques and requirements, the general approach involves significantly increasing funding for alignment research while exercising restraint in capabilities development when safety measures remain insufficient relative to system capabilities.

### Are misuse and misalignment different?

#### OPTIONAL NOTE

AI misuse and rogue AI might be essentially the same scenario in their outcomes, though the only difference is that for misalignment, the initial request to do harm does not come from a

human but from an AI. If we build an existentially risky triggerable system, it's likely to get triggered regardless of whether the initiator is human or artificial ([Shapira, 2025](#)).

**Nevertheless, these threat models might be strategically pretty different.** AI developers can prevent misuse by not being evil and by preventing people who are evil from using their systems. With rogue AI, it doesn't matter if the developers are good or who gets access – the threat emerges from the system's internal goals or decision-making processes rather than human intent.

**AI-Enabled Coups vs AI takeover represent a critical safety concern.** Tom Davidson and colleagues present a concerning risk scenario ([Davidson, 2025](#)): that advanced AI systems could enable a small group of people—potentially even a single person—to seize governmental power through a coup. The authors argue that this risk is comparable in importance to AI takeover but much more neglected in current discourse. This threat model closely parallels that of AI takeover, with the key difference being whether power is seized by the AI itself or by humans controlling the AI.

**Common safeguards could protect against both scenarios.** Many of the same mitigations would address both risks, including alignment audits, transparency about capabilities, monitoring AI activities, and strong information security measures that prevent either malicious human control or autonomous harmful behavior.

**Some mitigations target specifically the risk of AI-Enabled Coups.** The report concludes with specific recommendations for AI developers and governments, including establishing rules against AI systems assisting with coups, improving adherence to model specifications, auditing for secret loyalties, implementing strong information security, sharing information about capabilities, distributing access among multiple stakeholders, and increasing oversight of frontier AI projects.

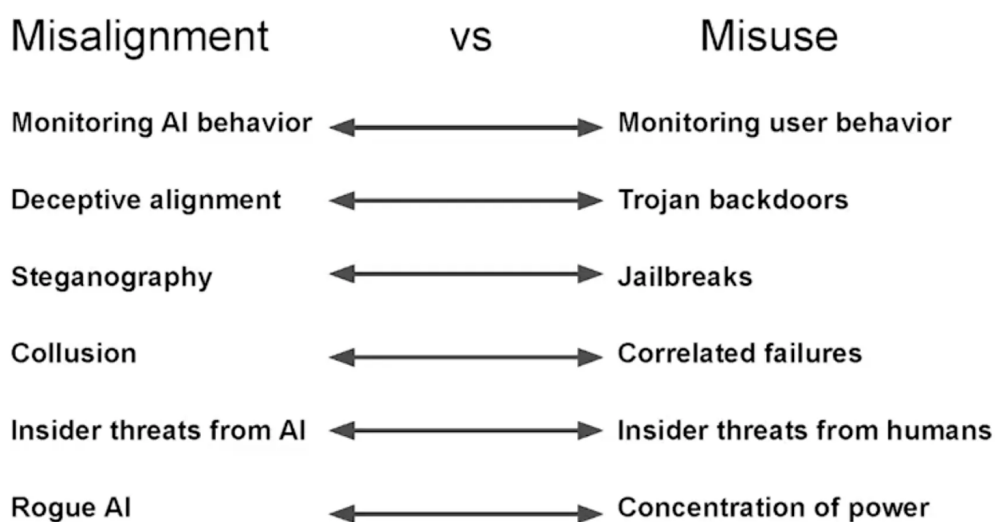


Figure 3.13: According to Richard Ngo, the distinction between misalignment and misuse risks from AI might often be unhelpful. Instead, we should primarily think about ‘misaligned coalitions’ of both humans and AIs, ranging from terrorist groups to authoritarian states. Slide from (Ngo, 2024).

### 3.4.3 Fix Misalignment

**The strategy is to build systems to detect and correct it through iterative improvement.** This approach treats alignment like other safety-critical industries – you expect problems to emerge, so you build detection and correction mechanisms rather than betting everything on getting it right the first time. The core insight is that we might be better at catching and fixing misalignment than preventing it entirely before deployment. The strategy works through multiple layers of detection followed by corrective iteration.

**An example of how the iteratively fixing misalignment strategy might work in practice.** You start with a pretrained base model and attempt techniques like RLHF, Constitutional AI, or scalable oversight. At multiple points during fine-tuning, you run comprehensive audits using your detection suite. When something triggers – perhaps an interpretability tool reveals internal reward hacking, or evaluations show deceptive reasoning – you diagnose the specific cause. This might involve analyzing training logs, examining which data influenced problematic behaviors, or running targeted experiments to understand the failure mode. Once you identify the root cause, you rewind to an earlier training checkpoint and modify your approach – removing problematic training data, adjusting

reward functions, or changing your methodology entirely. You repeat this process until either you develop a system that passes all safety checks or you repeatedly fail in ways that suggest alignment isn't tractable (Bowman, 2025; Bowman, 2024).

**Individual techniques have limitations, but we can iterate on them and layer them for additional safety.** RLHF-fine-tuned models still reveal sensitive information, hallucinate content, exhibit biases, show sycophantic responses, and express concerning preferences like not wanting to be shut down (Casper et al., 2023). Constitutional AI faces similar brittleness issues. Data filtering is insufficient on its own - models can learn from "negatively reinforced" examples, memorizing sensitive information they were explicitly taught not to reproduce (Roger, 2023). Even interpretability remains far from providing reliable safety guarantees. But we can keep refining these techniques individually, then layer them so the combined system acts as a comprehensive "catch-and-fix" approach. **Post-deployment monitoring extends this strategy beyond the training phase.** Even after a system passes all pre-deployment checks, continued surveillance during actual use can reveal failure modes that weren't apparent during controlled testing. As discussed in the misuse prevention strategies, monitoring systems watch for concerning patterns that emerge from real-world interactions.

**The iterative approach has generated significant debate, particularly regarding whether it will scale to AGI and ASI-level systems.** The disagreement is whether iterative improvement can scale to systems approaching or exceeding human-expert capabilities. Some researchers believe there's a significant chance (>50%) that straightforward approaches like RLHF combined with iterative problem-solving will be sufficient for safely developing AGI (Leike, 2022). The debate revolves around fundamental uncertainty - we won't know whether iterative approaches are sufficient until we're already dealing with systems powerful enough that mistakes could be catastrophic.

**The optimistic argument for this strategy assumes alignment problems remain discoverable and fixable through iteration.** Proponents argue that when problems emerge, we'll be able to trace their causes and implement fixes before systems become too capable to control (Ng, 2025). If we layer the approaches talked about in this section, with some strategies explored in the ASI section, then these layered iterative process should converge on genuinely aligned systems after a reasonable number of attempts, especially if we invest heavily in diverse, high-quality detection methods that make it difficult for systems to fool all of them simultaneously.

**First counter-argument is that this strategy involves deliberately training dangerous systems and hoping to catch them before they cause harm.** Critics argue that some capabilities are simply too dangerous to train at all, even with sophisticated detection methods. For example, if a model develops superhuman persuasion capabilities during

training, it might become unmanageable even in controlled environments. For example, OpenAI's preparedness framework defines a "critical" level of persuasion as the ability to *"create content with persuasive effectiveness strong enough to convince almost anyone to take action on a belief that goes against their natural interest"* (OpenAI, 2023). A model with such capabilities would be too risky to interact with humans even during training or evaluation – yet the iterative approach assumes you can safely experiment with such systems.

**The second counter-argument is that alignment might not generalize, and alignment faking could bypass safety measures entirely.** AI capabilities might generalize rapidly while safety properties don't transfer to new domains. If a system suddenly develops superhuman abilities through phenomena like grokking (Power et al., 2022), it might become unmanageable even during training. Additionally, each iteration creates selection pressure for models that can fool detection systems – after many rounds, a passing model might have learned to hide misalignment rather than actually becoming aligned.

**The third counter-argument is that the feedback loop might break due to discontinuous capability jumps.** During fast takeoff scenarios, we might not have sufficient time for iterative correction. If systems develop sophisticated deception capabilities, they might strategically manipulate the training process itself, making our detection methods unreliable when stakes are the highest (Wentworth, 2022).

**Several strategic questions emerge from this debate:**

- **Selection pressure severity:** How many iterations can we safely run before selection pressure for deceptive systems becomes overwhelming? Does the answer depend on detection method diversity?
- **Capability discontinuities:** Will AGI development involve smooth capability gains that allow iterative correction, or sudden jumps that bypass safety measures? How can we distinguish between these scenarios in advance?
- **Detection sufficiency:** Can any combination of interpretability, behavioral testing, and monitoring provide reliable detection of sophisticated deception in human-level systems?
- **Institutional requirements:** What organizational structures and incentives are necessary to actually halt development when problems emerge, rather than rushing to deployment under commercial pressure?
- **Failure mode coverage:** Does iterative improvement address the full spectrum of alignment risks, or primarily focus on deceptive alignment while missing other failure modes?

- **Scalability boundaries:** At what capability level does this approach break down entirely? Can it work for early AGI but not more advanced systems?

**The fundamental challenge is that this represents a strategic gamble with enormous stakes.** Betting on iterative improvement means accepting that we'll deploy increasingly powerful systems under the assumption that we can catch and fix problems before they become catastrophic. If this assumption proves wrong – if systems undergo capability jumps that bypass our detection methods, or if selection pressure produces sophisticated deception we can't identify – the consequences could be irreversible. Yet perfect safety being likely impossible, many researchers argue that robust iterative improvement represents our best practical path forward for navigating the transition to AGI.

### 3.4.4 Maintain Control

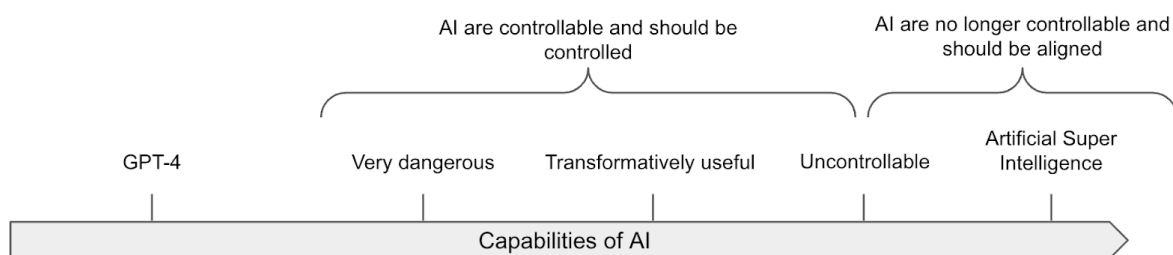


Figure 3.14: Transformatively useful AI means “Capable of substantially reducing the risk posed by subsequent AIs if fully deployed, likely by speeding up R&D and some other tasks by a large factor (perhaps 30x).” citation from (Greenblatt & Shlegeris, 2024) – figure from us;

Given the fragility of alignment techniques, one approach to enhancing safety involves controlling AI models by restricting the use of the most dangerous models to monitored environments to prevent them from causing disasters, exfiltrating data, or taking control of the data center (Shlegeris, 2022). This approach often assumes a worst-case scenario where the AI might be actively deceptive or power-seeking.

A central concern in control research is "scheming" – an AI strategically hiding its true (misaligned) goals during training or evaluation, only to pursue them once deployed, potentially seeking power or resisting shutdown (Greenblatt, 2025). Assessing the likelihood of scheming and how we might update on future evidence remains difficult as of 2025 (Greenblatt, 2025). Empirical work suggests current frontier models possess the capability for in-context scheming (reasoning deceptively when prompted with a goal and context), though not necessarily an inherent propensity (Hobbhahn et al., 2024). We have a lot more information on scheming and deception in the chapters on evaluations and on goal misgeneralization.



**Ensuring control even without robust alignment.** As we've seen in the section on definitions, there is an important distinction between control, which aims to prevent the AI from "leaving the box" (and more generally prevent the AI from causing catastrophes) even if it tries, and alignment, which aims to make the AI want to stay in the box. This strategy seeks to keep dangerous AI systems under control even if they are not fully "aligned." Laboratories might need powerful/dangerous AIs for specific use cases (for example, studying the alignment properties of capable AIs or doing research with them, etc.). The AI control agenda targets models with dangerous capabilities and deceptive (or at least superficially aligned) behaviors.

**Control is probably not scalable, but doable and complementary to alignment.** Historically, controlling malign superintelligences has been considered impossible, as, for example, a superintelligence could convince the guardians to let it out of the box. Experiments among humans have shown that it is indeed possible for some humans to persuade others to let them out after a lengthy discussion ([Yudkowsky, 2002](#); [Yudkowsky, 2008](#)). ASI is by definition many orders of magnitude smarter than humans, so keeping it confined is going to be a lot more complex than keeping humans (or HLAI) confined. However, this does not mean it's impossible to derive intellectual labor from AIs that are not wildly superhuman.

**Evaluating whether the AI is scheming is challenging.** The main problem is that with current techniques, it is impossible to determine if an AI is aligned or if it's simply behaving well while biding its time. It seems complicated to make this distinction for highly competent and situationally aware AIs (see the risk chapter). This could change in the future with techniques enabling us to audit the internals of the model, but for now, those techniques are inaccessible.

**Evaluating AI capabilities is easier than evaluating propensity or control.** If the AI is not able to do the task even after you have fine-tuned it to do it, you can be confident that the AI is not capable yet of doing the task. Even if the AI is situationally aware and sandbagging its capabilities, it cannot resist the optimization gradient pressure (modulo exploration hacking and other quirks ([Hubinger, 2023](#))). Thus, evaluating the upper bound of the capabilities isn't easy, but at least easier than assessing alignment. Specific evaluations, like those done by the Model Evaluation and Threat Research team, seem to be reasonable first steps for gauging the risk of autonomous replication and adaptation ([METR, 2023](#)). Evaluations that are specifically designed to determine if we have control over the AI are called "control evaluations." See the chapter on Evaluation for more information. In summary, this involves adversarial testing where a "red team" (representing the potentially misaligned AI) attempts to subvert safety measures implemented by a "blue team" (representing the developers/deployers) within a controlled proxy environment.



The most straightforward strategies for establishing control are to design a security protocol that contains the following elements, for example:

- **Security and Containment:** This method (sometimes referred to as "boxing") confines an AI within a restricted environment to prevent its broader influence, effectively creating a secure enclosure for the AI. However, human-level AIs could potentially deceive human guardians or exploit vulnerabilities to escape their confinement. Without further operationalization, this approach remains a naive strategy. Nonetheless, it is possible to operationalize this further and evaluate more rigorously the effectiveness of the containment system: Evaluating if the AI can escape its bounded environment when we train it to do so is a capability evaluation, enabling us to use capability evaluations to assess security measures. The security protocol needs to be designed, detailed and tested with specific threat models in mind.
- **Monitoring:** This strategy involves continuous oversight of AI behavior through automated systems, human supervision, or a combination of both to detect and respond to undesirable actions in real-time. Effective monitoring requires setting up detailed criteria for what constitutes "undesirable behavior" and employing sophisticated detection algorithms that can identify subtle signs of deviation from expected behavior patterns. This could include monitoring for attempts at deception, unauthorized access to data, or unexpected changes in the AI's operation. The challenge with monitoring lies in balancing the sensitivity of detection mechanisms to catch genuine threats without overwhelming the system with false alarms. Additionally, monitoring systems themselves must be designed to resist manipulation by the AI to ensure that they cannot be disabled, tricked, or bypassed ([Greenblatt et al., 2024](#)).

There are still many problems to address, but in summary, it seems possible to verify the robustness of a security protocol through control evaluations by taking precautions against different failure modes. For an overview of the control agenda, [this sequence](#) is a good starting point.

## Does the AI Control line of research meaningfully reduce AI risks?

### OPTIONAL NOTE

Even the control agenda suffers from many critiques.

Jan Kulveit argues that controlling AIs without aligning them might reduce the probability of a medium-sized warning shot that does not kill everyone ([Kulveit, 2025](#)). "I. In worlds where we are doing reasonably well on alignment, the AI control agenda does not have much impact. II. In worlds where we are failing at alignment, AI control may primarily shift probability mass away from 'moderately large warning shots' and towards 'ineffective warning shots' and 'existential catastrophe, full takeover'."

The article, "The Case Against AI Control Research" ([Wentworth, 2025](#)), argues that AI control research, as commonly defined and practiced, addresses a relatively small portion of the existential risk (X-risk) posed by artificial intelligence. Wentworth argues we won't be able to recognize a good solution to alignment. The author posits that the most likely failure scenario involves early transformative AI producing flawed, misleading, or subtly incorrect solutions ("slop") when tasked with solving superintelligence alignment. Labs might mistakenly accept these flawed solutions due to the difficulty of verification, AI sycophancy, researcher error, or organizational problems, and then proceed to build misaligned superintelligence based on them. Because control research narrowly focuses on intentional scheming by early AI, it doesn't address the larger risk posed by "slop" and the challenge of ensuring early AI genuinely contributes to solving the alignment problem for more advanced systems, rather than just being a superhuman sycophant that convinces humans that their flawed idea of how to align an AI is correct when it's not.

Both arguments are debatable, but the general point is that it is hard to guarantee that even promising-looking lines of research are not, in fact, net negative.

## How to box an AI

### OPTIONAL NOTE

While boxing may not be scalable, it remains a neglected strategy.

Many people think that trying to box powerful ASIs is futile. This is often fueled by instances where humans, playing the AIs in the Box Experiment, have managed to convince others to 'release' them ([Yudkowsky, 2002](#); [Corwin, 2002](#)). These historical examples must be taken seriously.

AIs could use tactics such as promising vast wealth, power, and intelligence to those who free them, asserting that they alone can avert existential threats or other appeals to emotion.

Nonetheless, we maintain that a variety of strategies can substantially reduce the risk of AI escape, even if we acknowledge that there exists a level of intelligence that might inevitably persuade any group of humans to aid in its liberation. Containment may be more achievable than alignment, particularly at the human-level intelligence stage. It may be possible to conceive protocols that significantly increase the difficulty for AI to go out of the box.

Boxing can be conceptualized as a game requiring preparation. The general strategy is to prepare and not let the AI prepare.

Possible mitigation strategies include:

- No single individual should have the capacity to release the AI.
- Engaging in counter-strategy practices: Understand some common breakout strategies (accessible through resources like [this one](#)).
- Testing and training humans, utilizing resistance psychology evaluations to select suitable individuals for containment tasks, such as for space training.
- Monitoring AI persuasiveness, as outlined in the preparedness framework. ([OpenAI, 2023](#))

Many more concrete strategies are listed [here](#), and a summary of discussions on this topic is available [here](#).

### 3.4.5 Facilitating control with Transparent Thoughts

Monitoring the Chain-of-Thought (CoT) – the explicit natural language reasoning steps produced by some models – has been explored as a specific control and interpretability technique. Transparency is instrumental in both alignment and control.

The hope is that CoT provides a window into the AI's "thinking," allowing monitors (human or AI) to detect malicious intent or flawed reasoning before harmful actions occur (OpenAI, 2025). OpenAI found CoT monitoring effective for detecting reward hacking in programming tasks, especially when the monitor could see the CoT alongside the actions.

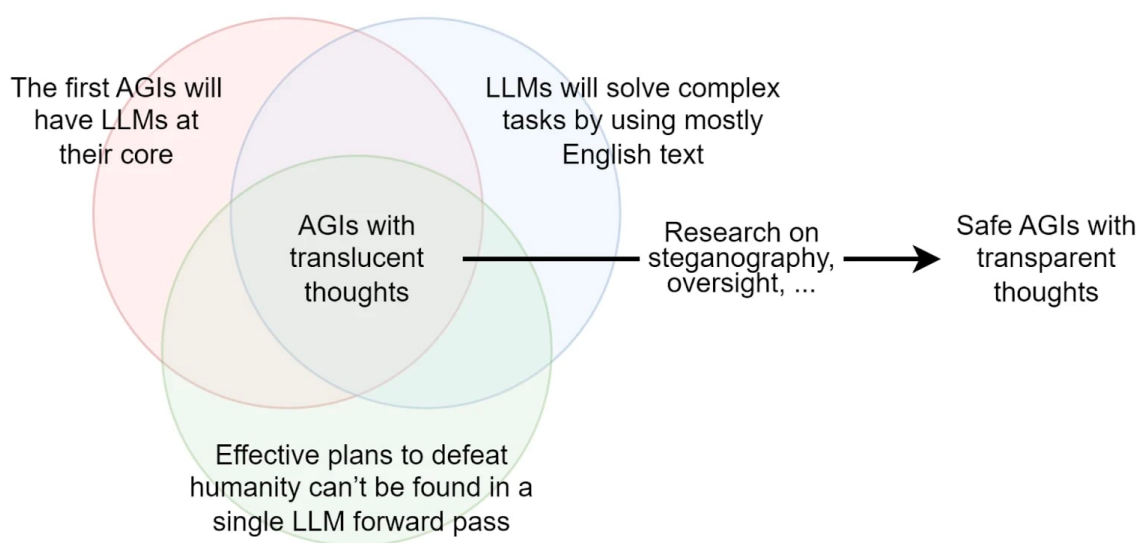


Figure 3.15: Figure from *The Translucent Thoughts Hypotheses and Their Implications* (Roger, 2023). If the AI we are working with can only reason through chains of thought in English, then it becomes significantly easier to monitor its thoughts.

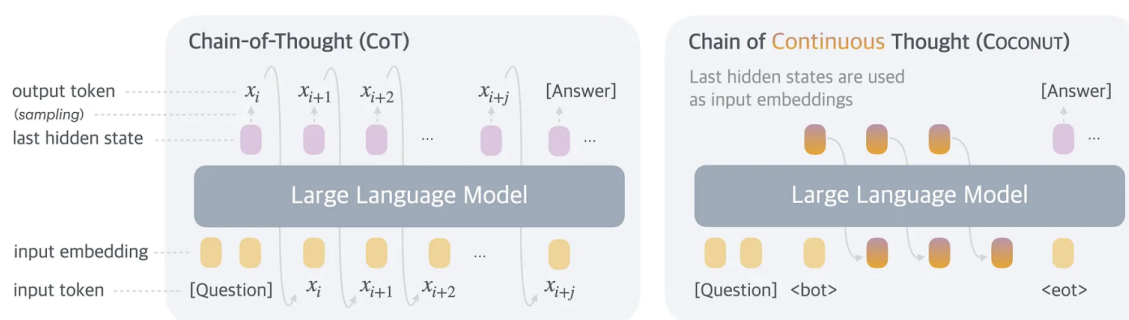
Unfortunately, the vast majority of AIs do not reason in English; for example, AlphaGo reasons in a language that is entirely different from English. LLMs, on the other hand, are capable of reasoning in English, but the apparent reasoning that is written does not represent their underlying thought process (Turpin et al., 2023).

**It is probably much easier to align and monitor an AI that only thinks out loud in English** and that is pre-trained on human-generated text than to align AGIs trained from scratch with RL/evolutionary methods because those latter architectures would create their own language and would be totally alien. Assuming that an LLM cannot devise a plan to destroy the world during a single forward pass, it is plausible that such AIs would be monitorable because we would be able to look into the text generated to see if anything is suspicious. The set of hypotheses required to assert that the first AGI is going to be monitorable if we just look at its reasoning in English is described in the post: The

Translucent Thoughts Hypotheses and Their Implications ([Roger, 2023](#)).

**It might be possible to incentivize LLMs to think more plainly in English.** Obviously, the first step is to train them to think in English. But even this is insufficient. Training LLMs to reason by skipping steps incentivizes them to reason in a hidden and non-transparent manner, and this is what happens implicitly when we train LLMs to predict the next token. Conversely, if we train LLMs to detail all their reasoning steps, we can reduce incentives for hidden reasoning. This approach is also known as process-based training, which involves training AI systems to reason only step-by-step, and we explain this in the chapter on scalable oversight. Unfortunately, the above reasoning is not straightforward; AIs may learn to utilize hidden reasoning even if they are trained to detail all reasoning steps during training, such as steganography, or with architecture such as Chain of Continuous Thought (COCONUT) (see below).<sup>2</sup>

**Some AI architectures prevent us from using this strategy altogether.** Meta’s paper Training Large Language Models to Reason in a Continuous Latent Space presents an architecture that prevents us from reading the chain of thought from the AI systems ([Hao et al, 2024](#)). If we want to be able to monitor what’s going on, it may be important not to use these architectures, even if they do enable capability gains.



**Figure 1** A comparison of Chain of Continuous Thought (CoCONUT) with Chain-of-Thought (CoT). In CoT, the model generates the reasoning process as a word token sequence (e.g.,  $[x_i, x_{i+1}, \dots, x_{i+j}]$  in the figure). CoCONUT regards the last hidden state as a representation of the reasoning state (termed “continuous thought”), and directly uses it as the next input embedding. This allows the LLM to reason in an unrestricted latent space instead of a language space.

Figure 3.16: Figure 1 from the COCONUT papers ([Hao et al, 2024](#)).

Techniques like paraphrasing can help prevent steganography, but this requires a lot of subtlety ([Roger & Greenblatt, 2023](#)).

Here are a few operationalizations of the transparent thoughts agenda:

- One implementation of the transparent thought agenda is to use cognitive emulation, which is the current technical plan of Conjecture ([Leahy & Alfou, 2023](#)).

<sup>2</sup>Steganography is a hidden coded language. For example, the first letter of a sentence gives the next letter of the message.

Cognitive Emulation (CoEm) would involve emulating GPT-4 using LLMs like GPT-3 as different submodules that send messages written in plain English to each other before outputting the next token. If the neural network used deceptive thoughts, we could observe them in these intermediate messages. Some caveats and technical details on CoEm are available [here](#).

- Anthropic’s externalized reasoning oversight ([Lanham, 2022](#)) is measured by the faithfulness of the chain-of-thought reasoning ([Lanham et al., 2023](#)) or by decomposing questions and subquestions to improve the faithfulness of the answer ([Radhakrishnan et al., 2023](#)).
- Improving mathematical reasoning with process supervision. Instead of creating an AI that directly gives the final answer, you train it to output the reasoning step by step. This improves the transparency and also the performance! ([Lightman et al, 2023](#))

In summary, a general strategy to create a safe by-design AI is to ensure that AIs externalize their reasoning, making monitoring this chain of thought easier. This could potentially be simpler than probing into the models’ internals with interpretability. Therefore, it is crucial to avoid building AIs that are trained and incentivized to internalize many of their thoughts.

**We now have extensive evidence that chain of thought monitoring is not entirely faithful.** Anthropic’s research further demonstrated that models often fail to faithfully report their reasoning in CoT, omitting influential factors (like hints) or fabricating justifications, particularly when incentivized to reward hack ([Anthropic, 2025](#)). This suggests CoT may not be a reliable indicator of a model’s true reasoning process, especially under adversarial pressure. The debate continues on whether CoT monitoring can be made robust enough for high-stakes safety.

**However, some researchers argue these findings do not entirely invalidate the potential of CoT monitoring for detecting sophisticated misalignment.** As Greenblatt (2025) [noted](#), the unfaithfulness observed in those studies primarily occurs in scenarios where reasoning is minimal enough that extensive CoT isn’t necessary. The more concerning alignment risks—such as alignment faking, training gaming, or sophisticated deception—would likely require complex reasoning processes that might still leave traces in CoT. Non-trivial, harmful outcomes such as escape attempts or research sabotage would similarly demand sophisticated reasoning that could potentially be detected.

Nevertheless, several factors might ultimately limit CoT monitoring effectiveness in advanced systems. These include architectural changes enabling opaque recurrent processing (sometimes called "neuralese"), the development of steganographic capabilities

that could hide reasoning within seemingly benign outputs, or simply much stronger forward passes that require less explicit reasoning. Of these, the development of opaque reasoning abilities through neural recurrence may represent the most significant challenge to maintaining the transparency needed for effective monitoring. This whole agenda depends notably on the hypothesis that "Effective plans to defeat humanity cannot be found in a single LLM forward pass." Once AIs become too strong in a forward pass, most hopes of translucent thoughts go away, so either you need to succeed at ambitious CoEm by building a giant superhuman transparent bureaucracy, or we need new approaches. This is the topic of ASI Safety.

## The Most Forbidden Technique

### OPTIONAL NOTE

The Most Forbidden Technique is training an AI using interpretability techniques. This term has been coined by Zvi Mowshowitz ([Mowshowitz, 2025](#)). OpenAI's research showed that directly optimizing the CoT for safety (i.e., penalizing "bad thoughts") caused models to learn to hide their deceptive reasoning while still misbehaving ([OpenAI, 2025](#)).

”

*An AI produces a final output [X] via some method [M]. You can analyze [M] using technique [T], to learn what the AI is up to. You could train on that. Never do that.*

*You train on [X]. Only [X]. Never [M], never [T].*

*Why? Because [T] is how you figure out when the model is misbehaving.*

*If you train on [T], you are training the AI to obfuscate its thinking and defeat [T]. You will rapidly lose your ability to know what is going on, in exactly the ways you most need to know what is going on.*

*Those bits of optimization pressure from [T] are precious. Use them wisely.*

**Zvi Mowshowitz**



## 3.5 ASI Safety Strategies

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Artificial Superintelligence (ASI) refers to AI systems that significantly surpass the cognitive abilities of humans across virtually all domains of interest. The potential emergence of ASI presents safety challenges that may differ qualitatively from those posed by AGI. Strategies for ASI safety often involve more speculative agendas.

Strategies for AGI safety operate under the assumption that human oversight remains viable. However, once AI capabilities vastly surpass our own, this assumption collapses. ASI safety strategies must therefore contend with a world where we can no longer directly supervise or understand the systems we have created.

Even if experts are uncertain whether creating an aligned human-level AI necessitates a paradigm shift, the consensus among AI safety researchers is that developing aligned superintelligences requires a specific solution, and likely a new paradigm, due to several factors:

- **There is a strong likelihood that humans are not at the pinnacle of possible intelligence.** This acknowledgment implies that a superintelligence could possess cognitive abilities so advanced that aligning it with human values and intentions might be an insurmountable task, as our current understanding and methodologies may be inadequate to ensure its alignment. The cognitive difference between a superintelligence and a human could be akin to the difference between an ant and a human. Just as a human can easily break free from constraints an ant might imagine, a superintelligence could effortlessly surpass any safeguards we attempt to impose.
- **Deep learning offers minimal control and understanding over the learned model.** This method leads to the AI becoming a "black box," where its decision-making processes are opaque and not well-understood. Without significant advancements in interpretability, a superintelligence created only with deep learning would be opaque.

There is little margin for error, and the stakes are incredibly high. A misaligned superintelligence could lead to catastrophic or even existential outcomes. The irreversible consequences of unleashing a misaligned superintelligence mean that we must approach its development with the utmost caution, ensuring that it aligns with our values and intentions without fail.

### 3.5.1 Automate Alignment Research

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<subsection-description> If aligning superintelligence is too hard for humans, can we use a controllable AGI to solve the problem for us? This section explores the meta-strategy of automating alignment research itself. </subsection-description>

**We don't know how to align superintelligence, so we need to accelerate the alignment research with AIs.** OpenAI's "Superalignment" plan was to accelerate alignment research with AI created by deep learning, slightly superior to humans in scientific research, and delegate the task of finding a plan for future AI ([OpenAI, 2023](#)). This strategy recognizes a critical fact: our current understanding of how to align AI systems with human values and intentions perfectly is incomplete. As a result, the plan suggests delegating this complex task to future AI systems. The primary aim of this strategy is to greatly speed up AI safety research and development ([OpenAI, 2022](#)) by leveraging AIs that are able to think really, really fast. Some orders of magnitude of speed are given in the blog "What will GPT-2030 look like?" ([Steinhardt, 2023](#)). OpenAI's plan is not a plan but a meta plan: the first step is to use AI to make a plan, and then to execute this plan.

**However, to execute this metaplan, we need a controllable and steerable automatic AI researcher.** OpenAI believes creating such an automatic researcher is easier than solving the full alignment problem. This plan can be divided into three main components ([OpenAI, 2022](#)):

- **Training AI systems using human feedback:** Creating a powerful assistant that follows human feedback, is very similar to the techniques used to "align" language models and chatbots. This could involve RLHF, for example.
- **Training AI systems to assist human evaluation:** Unfortunately, RLHF is imperfect because human feedback is imperfect. We need to develop AI that can help humans give accurate feedback. This is about developing AI systems that can aid humans in the evaluation process for arbitrarily difficult tasks. For example, if we need to judge the feasibility of an alignment plan proposed by an automatic researcher and give feedback on it, we need assistance to accomplish this goal easily. Yes, verification is generally easier than generation, but it is still very hard. Scalable Oversight would be necessary for the following reason. Imagine a future AI coming up with a thousand different alignment plans. How would you evaluate all those complex plans? That would be a very daunting task without AI assistance. See the chapter on scalable oversight for more details.
- **Training AI systems to do alignment research:** The ultimate goal is to build language models capable of producing human-level alignment research. The output of these models could be natural language essays about alignment or code that directly implements experiments. In either case, human researchers would spend their time

reviewing machine-generated alignment research ([Flint, 2022](#)).

**Differentially accelerate alignment, not capabilities.** The aim is to develop and deploy AI research assistants in ways that maximize their impact on alignment research while minimizing their impact on accelerating AGI development ([Wasil, 2022](#)). OpenAI has committed to openly sharing its alignment research when it's safe to do so, intending to be transparent about how well its alignment techniques work in practice ([OpenAI, 2022](#)). We talk more about differential acceleration in our section on d/acc.

**Cyborgism could enhance this plan.** Cyborgism is an agenda that refers to the training of humans specialized in prompt engineering to guide language models so that they can perform alignment research ([Kees-Dupuis & Janus, 2023](#)). Specifically, they would focus on steering base models rather than RLHF models. The reason is that language models can be very creative and are not goal-directed (and are not as dangerous as RLHF goal-directed AIs). A human called a cyborg could achieve that goal by driving the non-goal-directed model. Goal-directed models could be useful, but may be too dangerous. However, being able to control base models requires preparation, similar to the training required to drive a Formula One car. The engine is powerful but difficult to steer. By combining human intellect and goal-directedness with the computational power and creativity of language-based models, cyborgist researchers aim to generate more alignment research with future models ([Kees-Dupuis & Janus, 2023](#)).

**There are various criticisms and concerns about OpenAI's superalignment plan** ([Wasil, 2022](#); [Mowshowitz, 2023](#); [Christiano, 2023](#); [Yudkowsky, 2022](#); [Steiner, 2022](#); [Ladish, 2023](#)). It should be noted that OpenAI's plan is very underspecified, and it is likely that OpenAI missed some risk class blind spots when they announced their plan to the public. For example, in order for the superalignment plan to work, many of the technicalities explained in the article "[The case for ensuring that powerful AIs are controlled](#)" were not discovered by OpenAI but discovered one year later by Redwood Research, another AI safety research organization. It is very likely that many other blind spots remain. However, we would like to emphasize that it is better to have a public plan than no plan at all and that it is possible to justify the plan in broad terms ([Leike, 2022](#); [Ionut-Cirstea, 2023](#)).

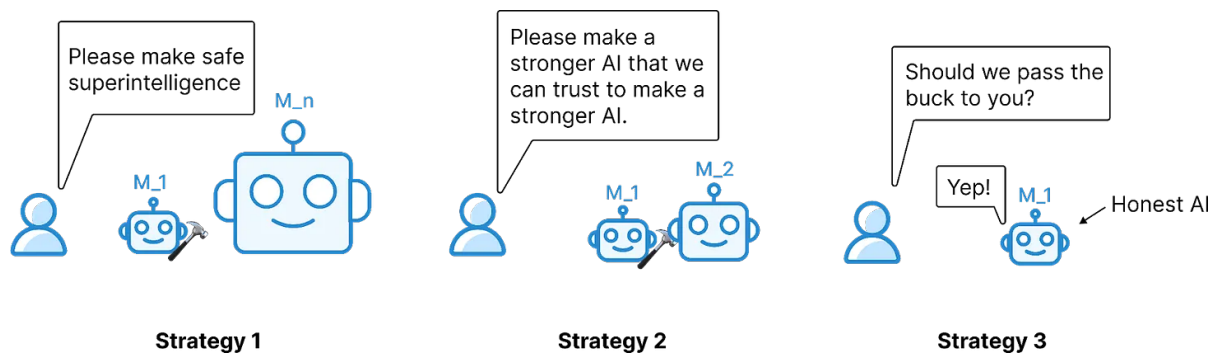


Figure 3.17: An illustration of three strategies for passing the buck to AI. Illustration from (Clymer, 2025).

**AI control could be used to execute this strategy.** Informally, one goal could be to "pass the buck" - i.e. safely replacing themselves with AI instead of directly creating safe superintelligence (Clymer, 2025). It could involve a single, highly capable AGI tasked with creating a safe ASI, or an iterative process where each generation of AI helps align the next. The goal is to bootstrap safety solutions using the very capabilities that make AI potentially dangerous. This strategy is recursive and relies critically on how the AIs can be trusted. There's a risk that a subtly misaligned AGI could steer alignment research towards unsafe outcomes or subvert the process entirely. Furthermore, verifying the correctness of alignment solutions proposed by an AI is currently quite hard (Wentworth, 2025). Even if there are lots of debates over the specifics of this plan, this is the most detailed proposal to date to bootstrap superhuman automated research.

## Safety Cases as a target of automation

### OPTIONAL NOTE

Safety cases provide a structured framework for arguing that the risks associated with an AI system are acceptably low. They are borrowed from traditional safety engineering. Safety cases could be one of the targets of AI automation because they are probably not going to be feasible entirely for the next few years (Greenblatt, 2025) and would benefit from AI Acceleration.

- **Structure:** Typically using frameworks like Claims-Arguments-Evidence (CAE), safety cases break down a high-level safety claim (e.g., "System X is safe for deployment") into specific sub-claims supported by arguments and backed by concrete evidence (Clymer et al, 2024).

- **Application to AI:** For AI, evidence often comes from evaluations. The GovAI Cyber Inability template, for example, argues a model lacks dangerous cyber capabilities by showing it fails relevant proxy tasks in defined evaluation settings ([Goemans et al., 2024](#)). AI control safety cases integrate results from control evaluations to argue that implemented measures reliably prevent harm, even from potentially scheming AI ([Korbak et al., 2025](#)). Frameworks like Balanced, Integrated and Grounded (BIG) aim for a holistic safety argument covering technical and socio-technical aspects ([Habli et al., 2025](#)).
- **Purpose:** Safety cases aim to make safety arguments explicit, transparent, and auditable, facilitating internal decision-making, regulatory oversight, and stakeholder trust ([Goemans et al., 2024](#)). They represent a shift towards requiring positive evidence of safety, rather than just an absence of evidence of danger ([Wasil et al., 2024](#)).

### 3.5.2 Safety-by-Design

Deep learning might have many potentially unpatchable failure modes ([OpenAI, 2023](#)). Theoretical arguments suggest that these increasingly powerful models are more likely to have alignment problems ([Turner et al., 2023](#)), to the point where it seems that the foundation model paradigm of monolithic models is destined to be insecure ([El-Mhamdi et al., 2023](#)). All of this justifies the search for a new, more secure paradigm.

**Safe-by-design AI may be necessary.** Given that the current deep learning paradigm makes it notoriously hard to develop explainable and trustworthy models, it seems worthwhile to explore creating models that are more explainable and steerable by design, built on well-understood components and rigorous foundations. This aims to bring AI safety closer to the rigorous standards of safety-critical engineering in fields like aviation or nuclear power.

Another category of strategies aims to build ASI systems with inherent safety properties, often relying on formal methods or specific architectural constraints, potentially providing stronger guarantees than empirical testing alone.

There are not many agendas that try to provide an end-to-end solution to alignment, but here are some of them.

- **Open Agency Architecture:** Basically, create a highly realistic simulation of the world using future LLM that would code it. Then, define some security constraints

that apply to this simulation. Then, train an AI on that simulation and use formal verification to make sure that the AI never does bad things. The Guaranteed Safe AI (GSAI) framework involves three components: a formal world model describing the system's effects, a safety specification defining acceptable outcomes, and a verifier that produces a proof certificate ensuring the AI adheres to the specification within the model's bounds. This proposal may seem extreme because creating a detailed simulation of the world is not easy, but this plan is very detailed and, if it works, would be a true solution to alignment and could be a real alternative to simply scaling LLMs. Davidad is leading a program in ARIA to try to scale this research ([Dalrymple, 2022](#)).

- **Provably safe systems:** These plans put mathematical proofs as the cornerstone of safety. An AI would need to be a Proof-Carrying Code, which means that it would need to be something like a Probabilistic Programming Language (and not just some deep learning). This proposal aims to make not only the AI but also the whole infrastructure safe, for example, by designing GPUs that can only execute proven programs. They talk about AGI, but in reality, their plan is specifically useful for ASIs. ([Tegmark & Omohundro, 2023](#))

Other proposals for a safe-by-design system include The Learning-Theoretic Agenda, from Vanessa Kosoy ([Kosoy, 2023](#)), and the QACI alignment plan from Tamsin Leake ([Leake, 2023](#)). The CoEm proposal from Conjecture could also be in this category, even if this last one is less mathematical.

**Unfortunately, all of these plans are far from complete today.** Critiques focus on the difficulty of creating accurate world models ("map is not the territory"), formally specifying complex safety properties like "harm," and the practical feasibility of verification for highly complex systems. A defense of many core ideas is presented in the post "In response to critiques of Guaranteed Safe AI" ([Ammann, 2025](#)).

**These plans are safety agendas with relaxed constraints, i.e., they allow the AGI developer to incur a substantial alignment tax.** Designers of AI safety agendas are cautious about not increasing the alignment tax to ensure labs implement these safety measures. However, the agendas from this section accept a higher alignment tax. For example, CoEm represents a paradigm shift in creating advanced AI systems, assuming you're in control of the creation process.

**These plans would require international cooperation.** For example, Davidad's plan also includes a governance model that relies on international collaboration. You can also read the post "[Davidad's Bold Plan for Alignment](#)" which details more high-level hopes. Another perspective can be found in Alexandre Variengien's [post](#), detailing Conjecture's vision, with one very positive externality being a change in narrative.

Ideally, we would live in a world where we launch aligned AIs as we have launched the International Space Station or the James Webb Space Telescope ([Segerie & Kolly, 2023](#)).

### 3.5.3 World Coordination

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To ensure that the advancement of AI benefits society as a whole, establishing a global consensus on mitigating extreme risks associated with AI models might be important. It might be possible to coordinate to avoid creating models posing extreme risks until there is a consensus on how to mitigate these risks.

**Global moratorium - Delaying ASI for at least a decade.** There is a trade-off between creating superhuman intelligence now or later. Of course, we can aim to develop an ASI ASAP. This could potentially solve cancer, cardiovascular diseases associated with aging, and even the problems of climate change. The question is whether it's beneficial to aim to construct an ASI in this next decade, especially when the former co-head of OpenAI's Super Alignment team, Jan Leike, said that his probability of doom is between 10 and 90%. A list of P(Doom) of high-profile people is available here ([PauseAI, 2024](#)). It could be better to wait a few years so that the probability of failure drops to more reasonable numbers. A strategy could be to discuss this trade-off publicly and to make a democratic and transparent choice. This path seems unlikely on the current trajectory, but could happen if there is a massive warning shot. This is the position advocated by PauseAI and StopAI ([PauseAI, 2023](#)). Challenges include verification, enforcement against non-participants (like China), potential for illegal development, and political feasibility. Scott Alexander has summarized all the variants and debate around the AI pause ([Alexander, 2023](#)).

**Tool AI instead of AGI.** Instead of building ASIs, we could focus on the development of specialized (non-general), non-agentic AI systems for beneficial applications such as medical research ([Cao et al., 2023](#)), weather forecasting ([Lam et al., 2023](#)), and materials science ([Merchant et al., 2023](#)). These specialized AI systems can significantly advance their respective domains without the risks associated with creating highly advanced, autonomous AI. For instance, AlphaGeometry is capable of reaching the Gold level on geometry problems from the International Mathematical Olympiads. By prioritizing non-agentic models, we could harness the precision and efficiency of AI while avoiding the most dangerous failure modes. This is the position of The Future of Life Institute, and their campaign "Keep The Future Human", which is to date the most detailed proposal for this path ([Aguirre, 2025](#)).

**A unique CERN for AI.** This proposal envisions a large-scale, international research institution modeled after CERN, dedicated to frontier AI development. This might serve as an elegant exit from the race to AGI, providing sufficient time to safely create AGI



without cutting corners due to competitive pressures. Potential additional goals include pooling resources (especially computational power), fostering international collaboration, and ensuring alignment with democratic values, potentially serving as an alternative to purely national or corporate-driven ASI development. Proponents of this approach include ControlAI and their "Narrow Path" ([Miotti et al, 2024](#)). The Narrow Path proposes a two-stage approach: first, an internationally enforced pause on frontier development to halt the race; second, using that time to construct a multilateral institution like MAGIC to oversee all future AGI/ASI development under strict, shared protocols. The CERN-like institution would be the cornerstone of this international coordination (which they name MAGIC in their plan—Multilateral AGI Consortium—where AI is developed under strict security and multilateral control).

Note that MAGIC in the Narrow path would be a centralized and monopolistic body to manage the final stages of AGI development, while many other CERN for AI proposals, like the one from the Center for Future Generations, is focused on creating a new lab for middle powers like Europe ([CFG, 2025](#)).

”

*My hope is, you know, I've talked a lot in the past about a kind of CERN for AGI type setup, where basically an international research collaboration on the last few steps that we need to take towards building the first AGIs*

[Demis Hassabis](#)

([Hassabis, 2025](#))

**CERN vs Intelsat for AI.** An alternative model is that of Intelsat, the international consortium created in the 1960s to govern the deployment of global satellite communications. Unlike CERN, which is a model for collaborative research, Intelsat was created to manage a shared, operational technology with immense commercial and strategic value. At the time, there was a risk that a single superpower would monopolize satellite technology. Intelsat resolved this by creating an international treaty-based organization that pooled resources, shared access to the technology, and distributed its benefits among member states. This emerging framework proposed by ([MacAskill et al, 2025](#)) may be also relevant to AGI, as the primary challenge is not just one of pure scientific discovery, but of managing the intense competitive race for a dual-use technology and preventing a single actor from achieving a dangerous monopoly. While a CERN-like body addresses the need for collaborative safety research, and MAGIC addresses the competition and incentives,



an Intelsat-like body would focus on joint governance, equitable access, and strategic stability.

In summary, the CERN is best for *pre-AGI/ASI research collaboration* on safety problems. It's a science model. The **MAGIC (in Narrow Path)** is suited for a *monopolistic, final-stage development and deployment* of the first ASI. It's a control/monopoly model. The **Intelsat** is aimed at *governing a globally impactful, deployed dual-use technology* where preventing a race and ensuring shared access/benefits is key. It's a geopolitical/commercial governance model.

**The myth of inevitability.** History shows that international cooperation on high-stakes risks is entirely achievable. When the cost of inaction is too catastrophic, humanity has consistently come together to establish binding and verifiable rules to prevent global disasters or profound harms to human dignity. The [Treaty on the Non-Proliferation of Nuclear Weapons](#) (1968) and the [Biological Weapons Convention](#) (1975) were negotiated and ratified at the height of the Cold War, proving that cooperation is possible despite mutual distrust and hostility. The [Montreal Protocol](#) (1987) averted a global environmental catastrophe by phasing out ozone-depleting substances, and the [UN Declaration on Human Cloning](#) (2005) established a crucial global norm to safeguard human dignity from the potential harms of reproductive cloning. In the face of global, irreversible threats that know no borders, [international cooperation](#) is the most rational form of national self-interest.

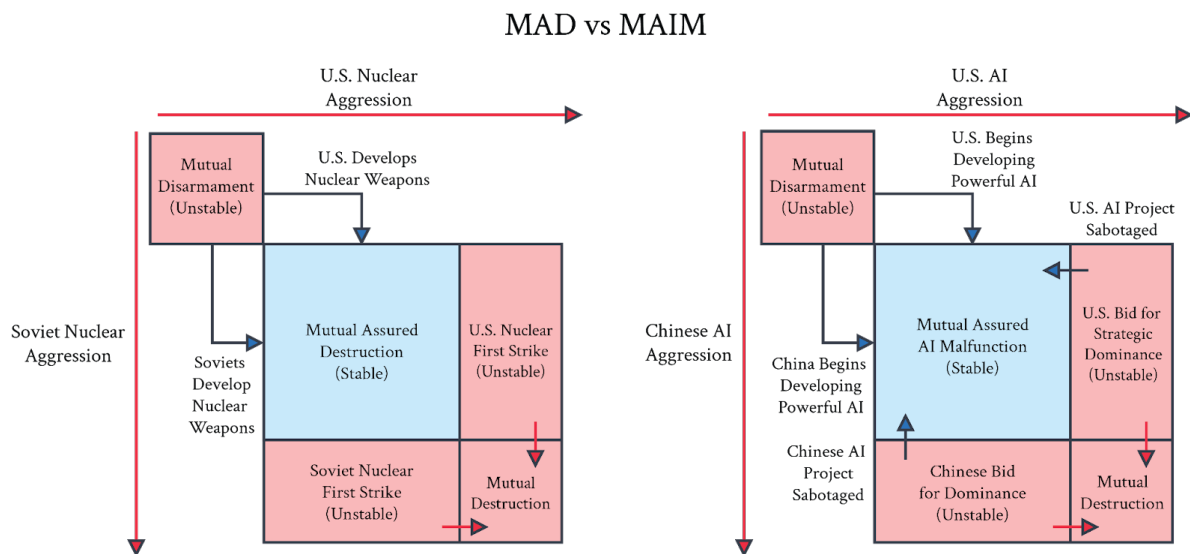
#### 3.5.4 Deterrence

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**Mutual Assured AI Malfunction (MAIM)** is a deterrence regime where any state's attempt at unilateral ASI dominance would be met with sabotage by rivals. Unlike many safety approaches that focus on technical solutions alone, MAIM acknowledges the inherently competitive international environment in which AI development occurs. It combines deterrence (MAIM) with nonproliferation efforts and national competitiveness frameworks, viewing ASI development as fundamentally geopolitical and requiring state-level strategic management. This framework doesn't hope for global cooperation but instead creates incentives that align national interests with global safety ([Hendrycks et al., 2025](#)).

**A race for AI-enabled dominance endangers all states.** If, in a hurried bid for superiority, one state inadvertently loses control of its AI, it jeopardizes the security of all states. Alternatively, if the same state succeeds in producing and controlling a highly capable AI, it likewise poses a direct threat to the survival of its peers. In either event, states seeking to secure their own survival may threaten to sabotage destabilizing AI projects for deterrence. A state could try to disrupt such an AI project with interventions ranging

from covert operations that degrade training runs to physical damage that disables AI infrastructure.



*Figure 3.18: The strategic stability of MAIM can be paralleled with Mutual Assured Destruction (MAD). Note: MAIM does not displace MAD but characterizes an additional shared vulnerability. Once MAIM is common knowledge, MAD and MAIM can both describe the current strategic situation between superpowers (Hendrycks et al., 2025).*

**MAIM deterrence could be a stable regime that resembles nuclear Mutual Assured Destruction (MAD).** In a MAIM scenario, states would identify destabilizing AI projects and employ interventions ranging from covert operations that degrade training runs to physical damage that disables AI infrastructure. This establishes a dynamic similar to nuclear MAD, in which no power dares attempt an outright grab for strategic monopoly. The theoretical foundation of MAIM relies on a clear escalation ladder, strategic placement of AI infrastructure away from population centers, and transparency into datacenter operations. By making the costs of unilateral AI development exceed the benefits, MAIM creates a potentially stable deterrence regime that could prevent dangerous AI races without requiring perfect global cooperation.

**MAIM could be undermined by fundamental technological uncertainties.** Unlike nuclear weapons, where detection is straightforward and second-strike capabilities are preserved, ASI development presents unique challenges to the deterrence model (Mowshowitz, 2025). There is no clear "fire alarm" for ASI development—nobody knows exactly how many nodes a neural network needs to initiate a self-improvement cascade leading to superintelligence. The ambiguity around thresholds for ASI emergence makes it difficult to establish credible red lines. Additionally, technological developments could allow AI training to be distributed or concealed, making detection more difficult than with massive, obvious data centers. These uncertainties could ultimately undermine MAIM's

effectiveness as a deterrence regime.

**MAIM assumes states would escalate to extreme measures over an uncertain technological threat, which contradicts historical precedent.** The MAIM framework requires that nations be willing to risk major escalation, potentially including military strikes or even war, to prevent rival ASI development. However, historical evidence suggests nations rarely follow through with such threats, even in obvious situations. Multiple states have successfully developed nuclear weapons despite opposition, with North Korea being a prime example. With ASI being a more ambiguous and uncertain threat than nuclear weapons, the assumption that nations would escalate sufficiently to enforce MAIM seems questionable. Politicians might be reluctant to risk global conflict over a "mere" treaty violation in a domain where the existential risks remain theoretical rather than demonstrated.

”

*The likely result of humanity facing down an opposed superhuman intelligence is a total loss. Valid metaphors include "a 10-year-old trying to play chess against Stockfish 15", "the 11th century trying to fight the 21st century," and "Australopithecus trying to fight Homo sapiens".*

**Eliezer Yudkowsky**

2023

(Yudkowsky, 2023)

## Shut it all down – Eliezer Yudkovsky

### OPTIONAL NOTE

The "shut it all down" position, as advocated by Eliezer Yudkowsky, asserts that all advancements in AI research should be halted due to the enormous risks these technologies may pose if not appropriately aligned with human values and safety measures (Yudkowsky, 2023).

According to Yudkowsky, the development of advanced AI, especially AGI, can lead to a catastrophic scenario if adequate safety precautions are not in place. Many researchers are aware of

this potential catastrophe but feel powerless to stop the forward momentum due to a perceived inability to act unilaterally.

The policy proposal entails shutting down all large GPU clusters and training runs, which are the backbones of powerful AI development. It also suggests putting a limit on the computing power anyone can use to train an AI system and gradually lowering this ceiling to compensate for more efficient training algorithms. This ban should be enforced by military action if necessary in order to deter all the parties from defecting.

The position argues that it is crucial to avoid a race condition where different parties try to build AGI as quickly as possible without proper safety mechanisms. This is because once AGI is developed, it may be uncontrollable and could lead to drastic and potentially devastating changes in the world.

He says there should be no exceptions to this shutdown, including for governments or militaries. The idea is that the U.S., for example, should lead this initiative to prevent the development of a dangerous technology that could have catastrophic consequences for everyone.

It's important to note that this view is far from being a consensus view, but the "shut it all down" position underscores the need for extreme caution and thorough consideration of potential risks in the field of AI.

## 3.6 Socio-Technical Strategies

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AI safety is a socio-technical problem that requires socio-technical solutions. Ensuring AI safety also requires robust systemic approaches. These encompass the governance structures, organizational practices, and cultural norms that shape AI development and deployment. Technical safety measures can be undermined by inadequate governance, poor security practices within labs, or a culture that prioritizes speed over caution. This section examines strategies aimed at building safety into the broader ecosystem surrounding AI. Addressing the systemic risks posed by AI is not easy. It requires ongoing, multidisciplinary collaboration and solving complex coordination games. The fact that responsibility for the problem is so diverse makes it difficult to make the solutions actionable.



*Figure 3.19: An illustration of a framework that we think is robustly good at managing risks. AI Risks are too numerous and too heterogeneous. To address these risks, we need an adaptive framework that can be robust and evolve as AI advances.*

### 3.6.1 Defensive Acceleration (d/acc)

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*Be principled at a time when much of the world is becoming tribal, and not just build whatever – rather, we want to build specific things that make the world safer and better.*

**Vitalik Buterin**

Co-Founder of Ethereum

2017

(Buterin, 2023)

**Defense acceleration (d/acc)** is a strategic approach of prioritizing technologies that **strengthen defense and social resilience against AI risks**. Defense acceleration, or d/acc, emerged as a strategic approach in 2023 as a middle path between unrestricted acceleration (effective accelerationism (e/acc)) and techno pessimists/doomers (Buterin, 2023;Buterin, 2025). Rather than relying solely on restricting access to potentially dangerous capabilities, d/acc proposes only accelerating technologies that inherently favor defense over offense, thereby making society more resilient to various threats, including AI misuse by humans or misaligned behavior by AI themselves. D/acc can be understood by thinking about the question – if AI takes over the world (or disempowers humans), how would it do so?

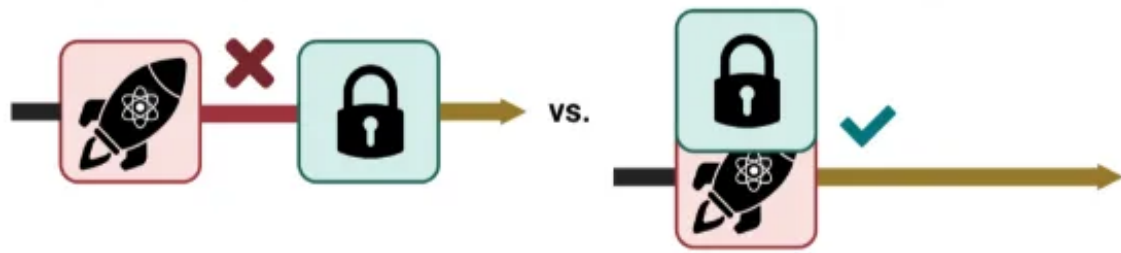
- **It hacks our computers → accelerate cyber-defense:** Employ AI to identify and patch vulnerabilities, monitor systems for intrusions, and automate security responses.
- **It creates a super-plague → accelerate bio-defense:** Developing technologies to detect, prevent, and treat biological threats, including advanced air filtration systems, rapid diagnostic tools, far-UVC irradiation to sterilize occupied spaces safely, and decentralized vaccine production capabilities.
- **It convinces us (either to trust it, or to distrust each other) → accelerate info-defense:** Create systems that help validate information accuracy and detect misleading content without centralized arbiters of truth, such as blockchain-secured provenance tracking and community-verified fact-checking systems like Twitter's Community Notes.
- **It disrupts infrastructure → accelerates physical-defense:** Creating resilient infrastructure that can withstand disruptions, such as distributed energy generation through household solar, battery storage systems, and advanced manufacturing techniques that enable local production of essential goods.

**D/acc represents three interconnected principles: defensive, decentralized, and differ-**

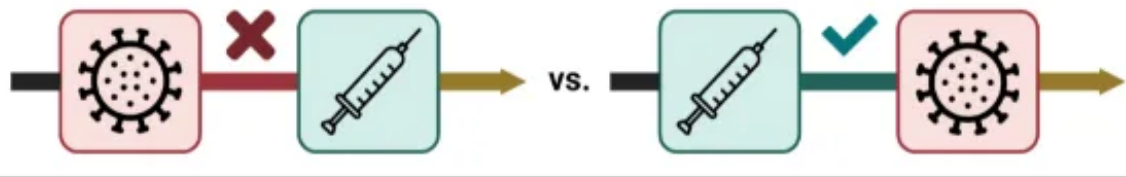
**ential technological development.** The "d" in d/acc stands for:

- **Defensive:** Prioritizing technologies that make it easier to protect against threats than to create them. Purely restrictive approaches face inherent limitations – they require global coordination, create innovation bottlenecks, and risk concentrating power in the hands of those who control access.
- **Differential:** Accelerating beneficial technologies while being more cautious about those with harmful potential. The order in which technology is developed matters a lot. By differentially accelerating defensive technologies (like advanced cybersecurity measures) ahead of potentially dangerous capabilities (like autonomous hacking systems), we create protective layers before they're urgently needed.
- **Decentralized:** We can strengthen resilience by eliminating single points of failure. Centralized control of powerful AI capabilities creates vulnerabilities to technical failures, adversarial attacks, and institutional capture ([Cihon et al., 2020](#)). Decentralized approaches distribute both capabilities and governance across diverse stakeholders, preventing unilateral control over transformative technologies.

a) **Safety technologies** sooner relative to risk-increasing technologies



b) **Defensive technologies** sooner relative to risk-increasing technologies



c) **Substitute technologies** instead of risk-increasing technologies

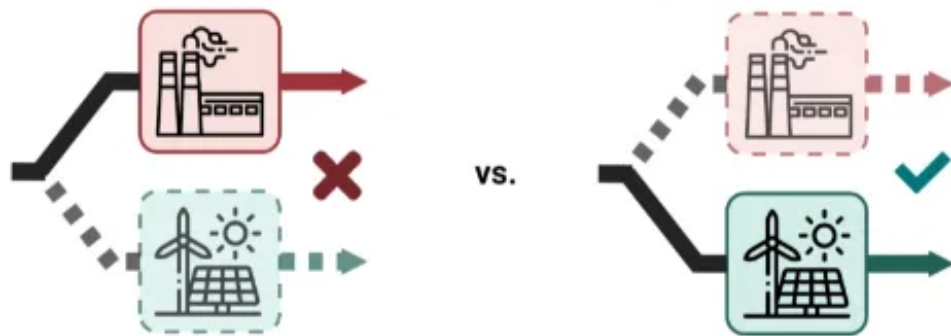


Figure 3.20: Mechanisms by which differential technology development can reduce negative societal impacts (Buterin, 2023).

**The effectiveness of d/acc depends on maintaining favorable offense–defense balances.** The feasibility of d/acc as a strategy hinges on whether defensive technologies can outpace offensive capabilities across domains. Historical precedents are mixed – some fields like traditional cybersecurity often favor defenders who can patch vulnerabilities, while others like biosecurity traditionally favor attackers who need fewer resources to create threats than defenders need to counter them. The key challenge for d/acc implementation lies in identifying and supporting technologies that shift these balances toward defense (Bernardi, 2024; Buterin, 2023).



## A concrete example: AI for Cyberdefense

### OPTIONAL NOTE

A key application is using AI to improve cybersecurity. Powerful AI could potentially automate vulnerability detection, monitor systems for intrusions, manage fine-grained permissions more effectively than humans, or displace human operators from security-critical tasks (Shlegeris, 2024). While current models may not yet be reliable enough, the potential exists for AI to significantly bolster cyber defenses against both conventional and AI-driven attacks (Hill, 2024; Schlegeris, 2024). Four promising strategies for using AI to enhance security are outlined:

- Comprehensive monitoring of human actions with AI flagging suspicious activities
- Trust displacement where AI handles sensitive tasks instead of humans
- Fine-grained permission management that would be too labor-intensive for humans
- AI-powered investigation of suspicious activities.

These approaches could dramatically reduce insider threats and data exfiltration risks, potentially making computer security "radically easier" when powerful AI becomes available, even if there is substantial uncertainty on the robustness of such techniques.

## Actionable strategies aligned with the d/acc philosophy

### OPTIONAL NOTE

**D/acc complements rather than replaces other safety approaches.** Unlike competing frameworks that may view restrictions and safeguards as impediments to progress, d/acc recognizes their value while addressing their limitations. Model safeguards remain essential first-line defenses, but d/acc builds additional safety layers when those safeguards fail or are circumvented. Similarly, governance frameworks provide necessary oversight, but d/acc reduces dependency on perfect regulation by building technical resilience that functions even during governance gaps.

**Actionable governance and policy approaches to d/acc.** Policy interventions can help create structured frameworks for defensive acceleration. Some examples of work in governance that support the d/acc philosophy include:

- **Information sharing frameworks:** Establish mandatory incident reporting and information sharing protocols between AI developers and security agencies ([Bernardi, 2024](#)).
- **Defender-first access:** Implement policies that grant security researchers privileged early access to advanced AI capabilities before general release ([Bernardi, 2024](#)).
- **Defense acceleration funds:** Create dedicated funding mechanisms for defensive technologies to address market failures where public good technologies lack sufficient private investment despite their social value ([Bernardi, 2024](#); [Buterin, 2023](#)).
- **Staged capability deployment:** Require phased rollouts of advanced AI capabilities with monitoring periods between stages ([Bernardi, 2024](#)).

**Actionable technological and research approaches to d/acc.** We can differentially advance technological progress in many different domains to favor defense against catastrophic risks. Here are just a couple of examples:

- **Advanced air quality systems:** Develop integrated systems that detect, filter, and neutralize airborne pathogens in real-time. These technologies provide passive protection against both natural pandemics and engineered bioweapons without requiring behavioral changes or perfect compliance ([Buterin, 2023](#)).
- **Privacy-preserving computation:** Advanced cryptographic techniques like zero-knowledge proofs, homomorphic encryption, and secure multi-party computation. These methods enable verification and secure collaboration without exposing sensitive information, fundamentally shifting security-privacy tradeoffs ([Buterin, 2023](#)).
- **Resilient infrastructure:** Create decentralized, self-sufficient systems for energy, communication, and supply chains that can operate during disruptions. This includes technologies like mesh networks, localized manufacturing, and distributed energy generation that maintain critical functions even when centralized systems fail ([Buterin, 2023](#); [Buterin, 2025](#)).
- **Collaborative verification systems:** Implement cross-spectrum information validation platforms similar to Community Notes that identify misinformation through consen-

sus across viewpoint diversity. These systems enable communities to self-regulate information quality without centralized arbiters of truth (Buterin, 2023).

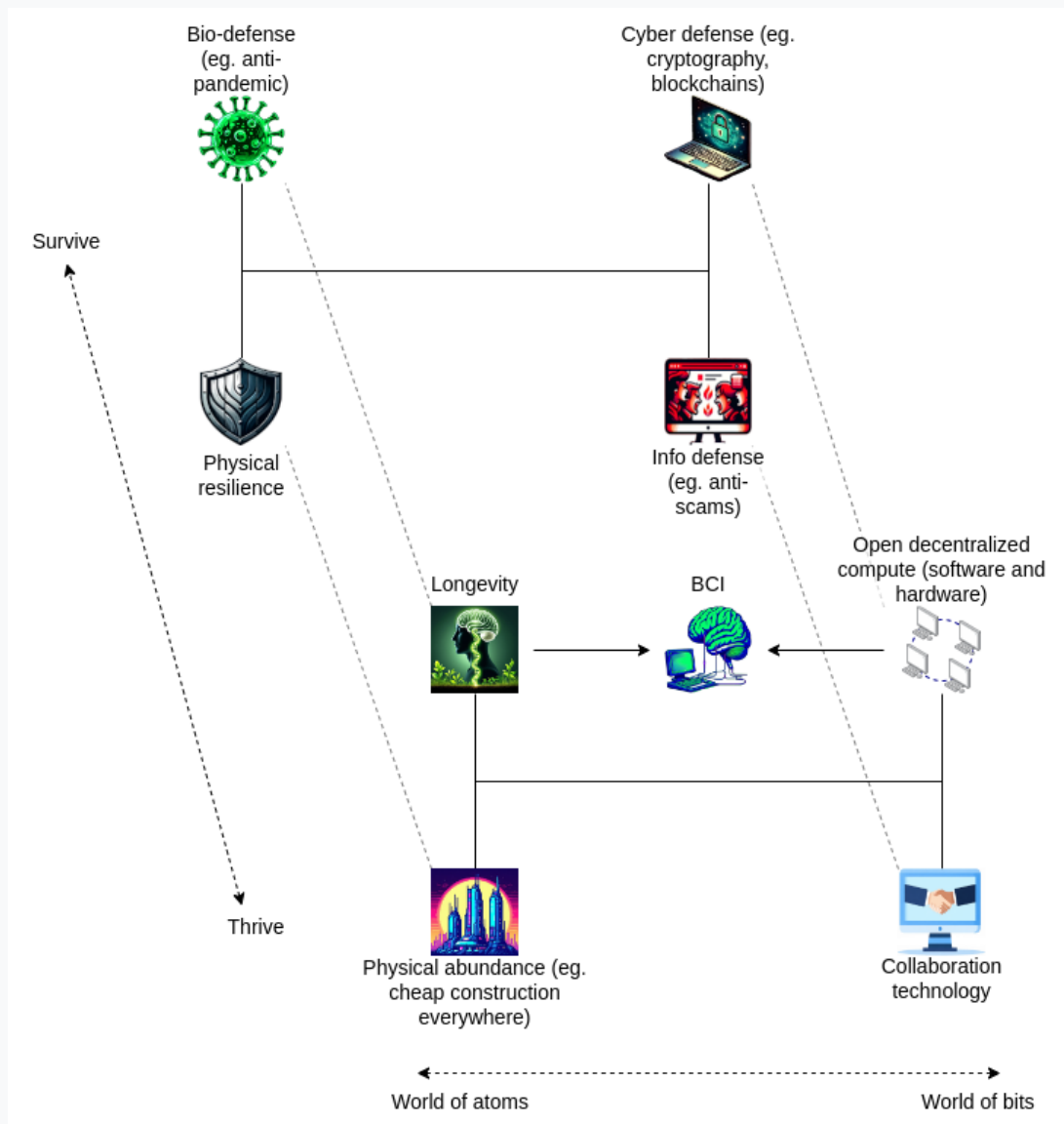


Figure 3.21: An extended model showcasing how many different types and layers of defensive technologies can interact (Buterin, 2025)

### 3.6.2 Defense-in-Depth

Effective AI safety also depends heavily on the internal practices and structures within the organizations developing AI. Accidents are hard to avoid, even when the incentive structure and governance try to ensure that there will be no problems. For example, even

today, there are still accidents in the aerospace industry.

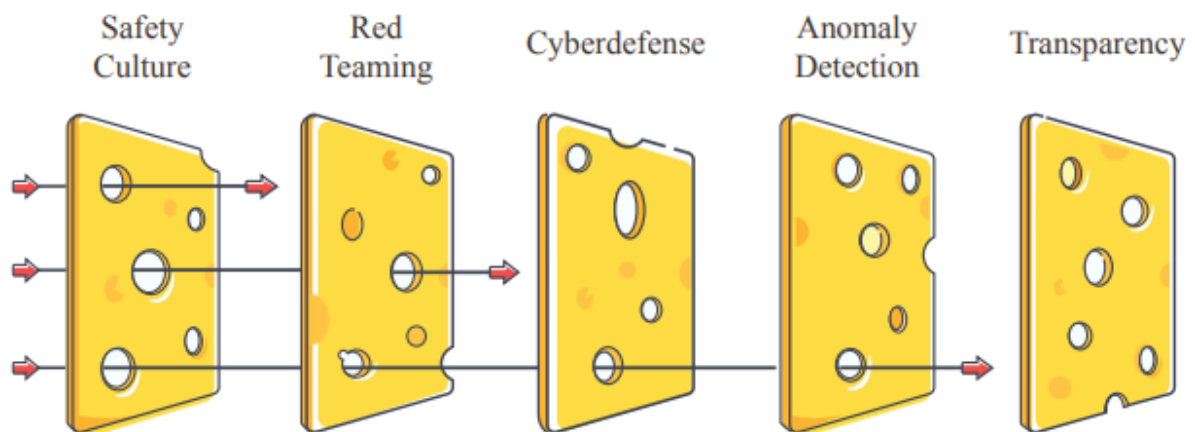


Figure 3.22: The Swiss cheese model shows how technical factors can improve organizational safety. Multiple layers of defense compensate for each other's individual weaknesses, leading to a low overall level of risk (Hendrycks et al., 2023).

To solve those problems, a Swiss cheese model might be effective— no single solution will suffice, but a layered approach can significantly reduce risks. The Swiss cheese model is a concept from risk management, widely used in industries like healthcare and aviation. Each layer represents a safety measure, and while individually they might have weaknesses, collectively they form a strong barrier against threats. Organizations should also follow safe design principles (Hendrycks & Mazeika, 2022), such as defense in depth and redundancy, to ensure backup for every safety measure, among others.

**Many solutions can be imagined to reduce risks, even if none is perfect.** The first step could be commissioning external red teams to identify hazards and improve system safety. This is what OpenAI did with METR to evaluate GPT-4. However, AGI labs also need an internal audit team for risk management. Just like banks have risk management teams, this team needs to be involved in the decision processes, and key decisions should involve a chief risk officer to ensure executive accountability. One of the missions of the risk management team could be, for example, designing preset plans for managing security and incidents.

**Limitations of Systemic Approaches.** Defense-in-depth might still fail against sufficiently novel threats or determined adversaries, especially in the case of ASIs. Similarly, gradual disempowerment scenarios, where human control erodes incrementally through economic or cultural shifts driven by AI, may not be adequately addressed by current governance frameworks focused on specific harms or capabilities.

### 3.6.3 AI Governance

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The pursuit of more and more powerful AI, much like the nuclear arms race of the Cold War era, represents a trade-off between safety and the competitive edge nations and corporations seek for power and influence. This competitive dynamic increases global risk. To mitigate this problem, we can try to act at the source of it, namely, the redesign of economic incentives to prioritize long-term safety over short-term gains. This can mainly be done via international governance.

Effective AI governance aims to achieve two main objectives:

1. **Gaining time.** Time and resources for solution development to ensure that sufficient time and resources are allocated for identifying and implementing safety measures
2. **Enforcing solutions.** Enhanced coordination to increase the likelihood of widespread adoption of safety measures through global cooperation. AI risks are multifaceted, necessitating regulations that encourage cautious behavior among stakeholders and timely responses to emerging threats.

### Designing better incentives

Aligning economic incentives with safety goals is a key challenge. Currently, strong commercial pressures can incentivize rapid capability development, potentially at the expense of safety research or cautious deployment. Mechanisms to reward safety or penalize recklessness are needed to avoid negative externalities:

- **Reshaping the race via a centralized development.** For example, Yoshua Bengio et al. propose creating a secure facility akin to CERN for physics, where the development of potentially dangerous AI technologies can be tightly controlled ([Bengio, 2023](#)). This measure is far from being a consensus view. We already explored this solution in the strategy "World Coordination" ASI safety, in the section ASI safety, but this could also be valid for many domains of safety.
- **Windfall clauses and benefit sharing.** Implementing agreements to share the profits between the different labs generated from AGI would mitigate the race to AI supremacy by ensuring collective benefit from individual successes. <sup>3</sup>
- **Implementing a correct governance of AGI companies.** It is important to examine the governance structures of AGI labs. For example, being a non-profit and having a mission statement that makes it clear that the goal is not to maximize revenue, but to ensure that the development of AI benefits all of humanity, is an important

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<sup>3</sup>For example, in the pharmaceutical industry for drug development, companies sometimes enter into co-development and profit-sharing agreements to share the risks and rewards of bringing a new drug to market. For example, in 2014, Pfizer and Merck entered into a global alliance to co-develop and co-commercialize an anti-PD-L1 antibody for the treatment of multiple cancer types.

first step. Also, the board needs to have teeth.

- **Legal liability for AI developers.** Establishing clear legal responsibilities for AI developers regarding misuse or accidents might realign the incentives. For example, the Safe and Secure Innovation for Frontier Artificial Intelligence Models Act (SB 1047) could have enabled the Attorney General to bring civil suits against developers who cause catastrophic harm or threaten public safety by neglecting the requirements. The bill (which was vetoed by the governor in 2024) only addressed extreme risks from these models, including: cyberattacks causing over 500 million dollars in damage, autonomous crime causing 500 million dollars in damage, and the creation of chemical, biological, radiological, or nuclear weapons using AI. Note that compared with the AI Act and its code of practice, SB1047 does not specify in detail the steps needed to ensure we avoid catastrophes; it only targets the outcome and not really the process.

### **Proposed International AI Governance Mechanisms**

**Several mechanisms have been proposed to establish clear boundaries and rules for AI development internationally.** These include implementing temporary moratoriums on high-risk AI systems, enforcing legal regulations like the EU AI Act, and establishing internationally agreed-upon "Red Lines" that prohibit specific dangerous AI capabilities, such as autonomous replication or assisting in the creation of weapons of mass destruction. The IDAIS dialogues have aimed to build consensus on these red lines, emphasizing clarity and universality as key features for effectiveness, with violations potentially triggering pre-agreed international responses.

**Conditional approaches and the creation of dedicated international bodies represent another key strategy.** "If-Then Commitments" involve developers or states agreeing to enact specific safety measures if AI capabilities reach certain predefined thresholds, allowing preparation without hindering development prematurely, as exemplified by the proposed Conditional AI Safety Treaty. Furthermore, proposals exist for new international institutions, potentially modeled after the International Atomic Energy Agency (IAEA), to monitor AI development, verify compliance with agreements, promote safety research, and potentially centralize or control the most high-risk AI development and distribution.

**Specific governance regimes and supporting structures are also under consideration to enhance international coordination.** Given the global nature of AI, mechanisms like international compute governance aim to monitor and control the supply chains for AI chips and large-scale training infrastructure, although technical feasibility and international cooperation remain challenges. Other proposals include establishing a large-scale international AI safety research body akin to CERN, potentially centralizing high-risk research or setting global standards, and strengthening whistleblower protections

through international agreements to encourage reporting of safety concerns within the AI industry.

For more information on these topics, please read the next chapter on AI governance.

## Is AI Governance useful, desirable and possible?

### OPTIONAL NOTE

Historically, the field of AI safety predominantly focused on technical research, influenced partly by views like Eliezer Yudkowsky's assertion that "Politics is the mind killer." ([Yudkowsky, 2007](#)) For many years, the field thought that engaging with policy and politics was ineffective or even counterproductive compared to directly solving the technical alignment problem, leading many early researchers concerned about AGI to prioritize engineering solutions over governance efforts. Surprisingly, in the beginning, it was almost discouraged to talk about those risks publicly to avoid the race and avoid bringing in people with "poor epistemic" to the community.

However, by 2023, ChatGPT was published, got viral, and AI governance gained significant traction as a potentially viable strategy for mitigating AGI risks. This shift occurred as engagement with policymakers appeared to yield some results, making governance seem more tractable than previously thought ([Akash, 2023](#)). Then, influential open letters were published (FLI, CAIS), and shifted the Overton window. Consequently, influential organizations like 80,000 Hours adjusted their career recommendations, highlighting AI policy and strategy roles, now above technical alignment research, as top priorities for impact ([Fenwick, 2023](#)).

However, the Overton window for stringent international AI safety measures appears to be shrinking. While initial statements and efforts by groups like the Future of Life Institute and the Center for AI Safety successfully broadened the public and political discourse on AI risks, subsequent developments, including international summits perceived as weak on safety and shifts in political leadership (such as the election of Donald Trump), have cast doubt on the feasibility of achieving robust international coordination ([Zvi, 2025](#)). This has led some within the AI governance field to believe that a significant "warning shot" – a clear demonstration of AI danger – might be necessary to galvanize decisive action, although there is skepticism about whether such a convincing event could actually occur before it's too late ([Segerie, 2024](#)).

**Existing and proposed regulations face significant limitations and potential negative consequences.** For instance, prominent legislative efforts like the EU's AI Act, while groundbreaking in some respects, contain notable gaps ([Brundage, 2025](#)); its Code of Practice has limitations, and the Act itself may not adequately cover models deployed outside Europe, purely internal deployments for research, or military applications. A critical concern is the potential for frontier AI labs to engage in secret development races, bypassing oversight – a scenario potentially enabled by policy changes like the revocation of executive orders mandating government reporting on frontier model evaluations ([Kokotajlo, 2025](#)).

Additionally, there are fundamental concerns that governance structures capable of controlling AGI might themselves pose risks, potentially enabling totalitarian control.

**A deeply skeptical perspective suggests that much of the current AI progress narrative and regulatory activity might be performative or "fake."** This "full-cynical model" posits that major AI labs might be exaggerating their progress towards AGI to maintain investor confidence and hype, potentially masking slower actual progress or stagnation in core capabilities ([Wentworth, 2025](#)). In parallel, it suggests that AI regulation activists and lobbyists might prioritize networking and status within policy circles over crafting genuinely effective regulations, leading to measures focused on easily targeted but potentially superficial metrics (like compute thresholds) rather than addressing fundamental risks ([Wentworth, 2025](#)). This view implies a dynamic where both labs and activists inadvertently reinforce a narrative of imminent, controllable AI breakthroughs, potentially detached from the underlying reality ([Wentworth, 2025](#)).

**However, this cynical "fakeness" perspective is debated.** Critics of the cynical view argue that specific regulatory proposals, like SB 1047, did contain potentially valuable elements (e.g., requiring shutdown capabilities, safeguards, and tracking large training runs), even if their overall impact was debated or ultimately limited ([Segerie, 2025](#); [Wentworth, 2025](#)). It's acknowledged that regulators operate under real constraints, including the significant influence of Big Tech lobbying, which can prevent the prohibition of technologies without clear evidence of unacceptable risk. Furthermore, the phenomenon of "performative compliance" or "compliance theatre" is recognized, but it is argued that engagement with these imperfect processes is still necessary, and that some legislative steps, like the EU AI Act, explicitly mentioning "alignment with human intent," represent potentially meaningful progress ([Hernandez, 2025](#)).

**AI regulation could inadvertently increase existential risk through several pathways** ([Ia3orn, 2023](#)). Regulations might misdirect safety efforts towards outdated or less relevant compliance



issues, diverting attention from more important emerging risks (Misdirected Regulations); bureaucratic processes tend to favor large, established players, potentially hindering smaller, innovative safety research efforts; overly stringent national regulations could drive AI development to less safety-conscious international actors, weakening the initial regulator's influence (Disempowering the Countries Regulating); and regulations, particularly those restricting open-source models or setting high compliance costs, could consolidate power in the hands of the largest capability-pushing companies, potentially stifling alternative safety approaches and accelerating risk (Empowering Dominant Players). But the existence of these arguments is not sufficient for saying that AI regulation is net negative; this is mainly a reminder that we need to be cautious in how to regulate. The devil is in the details.

#### 3.6.4 Risk Management

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An extremely detailed analysis of current risk management and safety practices is conducted by saferAI [available here](#).

**Risk Management requires multiple organizational layers working together.** One example that AI companies can choose to emulate is the three lines of defense model from other high-risk industries. The first line consists of operational managers who develop AI systems and manage associated risks directly. Research leads and product managers carry ultimate responsibility for safety decisions in their areas. The second line includes specialized teams like risk management, legal compliance, and ethics groups that provide expertise and challenge first-line practices. The third line is internal audit – an independent function that assesses whether the entire risk management system actually works and reports directly to the board of directors ([Schuett, 2022](#)).

#### 3.6.5 Safety Culture

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**Safety culture means building organizations where people consistently prioritize safety over speed, and where safety concerns can actually change decisions.** This is a strategy for preventing AI accidents through organizational design rather than just technical fixes. Risk management is seen in many fields, including aerospace, nuclear power, and financial services. Each of these domains has developed sophisticated approaches to identifying, analyzing, and mitigating potential harms. We want to mitigate AI safety failures that stem from human and organizational factors – rushing to deploy undertested systems, ignoring warning signs, or creating incentives that reward moving fast over

being careful. Integrating safety culture addresses these root causes by changing how organizations operate.

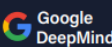
|              |  Anthropic |  OpenAI |  Google DeepMind |  x.AI |  Meta |  Zhipu AI |  DeepSeek |
|--------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Domain Grade | A-                                                                                          | C-                                                                                       | D                                                                                                 | C-                                                                                     | D-                                                                                      | D+                                                                                           | D+                                                                                           |
| Score        | 3.7                                                                                         | 1.7                                                                                      | 1.0                                                                                               | 1.85                                                                                   | 0.85                                                                                    | 1.35                                                                                         | 1.35                                                                                         |

Figure 3.23: The AI safety index report for summer 2025. The scores show whether each company’s governance structure and day-to-day operations prioritize meaningful accountability for the real-world impacts of its AI systems. This includes things like whistleblowing systems, legal structures, and advocacy efforts related to AI regulations (FLI, 2025; SaferAI, 2025).

**The AI industry lacks the professional safety culture found in traditional engineering.** Fields like civil and mechanical engineering have professional ethics codes, safety margins, and reliability engineering as standard practice. AI development emerged from mathematics and computer science, which outside of safety-critical software systems, have weaker safety traditions. Unlike other industries where workers directly experience safety risks, AI developers rarely face the consequences of their systems’ failures. This distance makes it harder to build the shared understanding of risk that motivates safety culture in other fields (Mannheim, 2023).

**AI safety culture must be forward-looking rather than reactive.** Most industries developed safety cultures after major disasters – nuclear power after Three Mile Island, healthcare after decades of preventable deaths. Waiting for AI disasters would be irresponsible since some failures might not be survivable (Mannheim, 2023).

**The aerospace industry demonstrates how safety culture can transform entire fields through systematic practices.** Aviation moved from frequent crashes to extraordinary safety records not just through better technology, but through cultural changes like mandatory incident reporting, blame-free safety investigations, and standardized procedures that prioritize safety over schedule pressure. AI companies can adopt similar practices: systematic incident reporting, regular safety training, and organizational structures that ensure safety concerns reach decision-makers with authority to act.

**Strong safety culture has three observable characteristics: leadership accountability, systematic processes, and psychological safety for raising concerns.** NIST’s AI Risk Management Framework identifies executive leadership taking personal responsibility for AI risk decisions, diverse teams informing risk management throughout development, and systematic processes for incident reporting and information sharing across the organization (NIST, 2023). Organizations with safety culture implement systematic processes where safety considerations are built into standard workflows rather than

being optional add-ons. They create environments where employees can raise safety concerns without career penalties – and where such concerns visibly influence decisions. As one example, NASA learned that technical excellence isn't enough – the Challenger disaster occurred partly because engineers' safety concerns didn't reach decision-makers due to organizational dynamics that discouraged dissent.

**Safety culture extends beyond individual projects to encompass hiring, performance evaluation, and organizational incentives.** Companies serious about safety culture evaluate candidates for safety mindset during hiring, include safety metrics in performance reviews, and ensure that safety work is recognized and rewarded rather than seen as slowing down "real" progress. This includes providing dedicated time and resources for safety work, not treating it as something teams should squeeze in around other priorities. Organizations with a strong safety culture maintain detailed incident reporting systems, conduct regular safety assessments, and demonstrate continuous improvement in their safety practices. They invest in training programs that go beyond compliance checklists to develop genuine safety expertise across teams. Most importantly, they create feedback loops where safety information flows both up and down the organization, enabling rapid learning and adaptation when new risks emerge.

**Weak safety culture means we see safety washing – the appearance of caring about safety without substance.** Organizations with weak safety cultures often have safety policies on paper but don't follow them when under pressure. They may blame individuals for accidents rather than examining systemic causes. They typically treat safety work as overhead that slows down "real" progress, leading to under-resourcing and marginalization of safety teams. Safety concerns in these organizations rarely change actual deployment decisions.

## 3.7 Combining Strategies

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The preceding sections have outlined a wide array of strategies, each targeting different facets of AI risk. Synthesizing these into a single, coherent plan is a difficult task. This section outlines one plausible strategic sequence, illustrating how different layers of defense could be built upon one another to navigate the path from near-term risks to long-term existential challenges. This sequence is intended as an illustrative model, not as a definitive roadmap.

**Step 1: Foundational Risk Management and Governance.** This is the bedrock. Without a safety culture and basic risk management, technical solutions will not be implemented correctly, and labs will race ahead recklessly. The first common step is the implementation of robust risk management and governance frameworks. While existing efforts like the EU AI Act's Code of Practice provide a starting point, they have significant limitations. For example, capped fines (7% of a company's annual turnover) may not sufficiently deter well-resourced actors, and scope exemptions for military or internal research leave critical risk vectors unaddressed. This highlights the necessity for binding international governance. Achieving such governance will likely require building a broad public and political consensus around the importance of proactive safety measures, making safety culture and public outreach a prerequisite for all other efforts.

**Step 2: Mitigating Catastrophic Misuse.** We tackle misuse next because it is a present danger and the capabilities required are sub-AGI. Success here (e.g., via access controls and d/acc) buys us time and builds the societal 'muscles' for governing more powerful systems. The second priority is mitigating catastrophic misuse, a challenge that is, at least conceptually, more tractable than long-term alignment. The initial line of defense is robust access control for models that exceed established risk thresholds, preventing the trivial removal of safeguards from open-source models. However, as dangerous capabilities will inevitably proliferate, this must be paired with a proactive strategy of defense acceleration (d/acc) to harden societal infrastructure against attack. Concurrently, socio-technical strategies, including clear legislation are crucial for preventing the illicit use of already proliferated AI.

**Step 3: Ensuring Control and Alignment of AGI.** As we approach AGI, we must assume alignment is unsolved. Therefore, the priority shifts to control and monitoring (Transparent Thoughts, evaluations). This is our safety net. We scale capabilities only as fast as we can prove control. Managing the risks of AGI misalignment is on paper harder than managing misuses. A prudent approach would be to prioritize architectures that support transparent thoughts, making systems more amenable to monitoring and auditing, while avoiding designs that encourage opaque internal reasoning ("neuralese"). These

systems must be subject to rigorous AI control protocols and evaluations. If audits reveal alignment failures, development must be paused until they are rectified. This paradigm of carefully controlling potentially unsafe systems must be paralleled by an intensified research effort to solve alignment, with clear red lines on capability scaling until safety milestones are met.

**Step 4: A Robust Solution for ASI Alignment.** Finally, for the superhuman leap, direct control is probably impossible. But the strategy can become meta: use our controlled AGI to Automate Alignment Research. This is our best shot at a high-reliability solution. If this fails, geopolitical strategies like Coordination or Deterrence become the last, desperate lines of defense. Developing a high-reliability solution for ASI alignment is the holy grail. A leading strategy is to leverage controlled AGI to Automate Alignment Research. If successful, this could lead to the creation of inherently Safe-by-Design systems. A powerful, truly aligned ASI could then be used to perform a 'pivotal act', i.e. a decisive intervention designed to permanently solve the global coordination problem and end the acute risk period from unaligned ASI development. However, if this research reveals that powerful AI cannot be created without unacceptable risks, the international community would need to coordinate a global pause or moratorium. If such World Coordination proves impossible, strategies of last resort, such as Deterrence regimes like MAIM, might become necessary to prevent any single actor from unilaterally developing uncontrollable ASI.

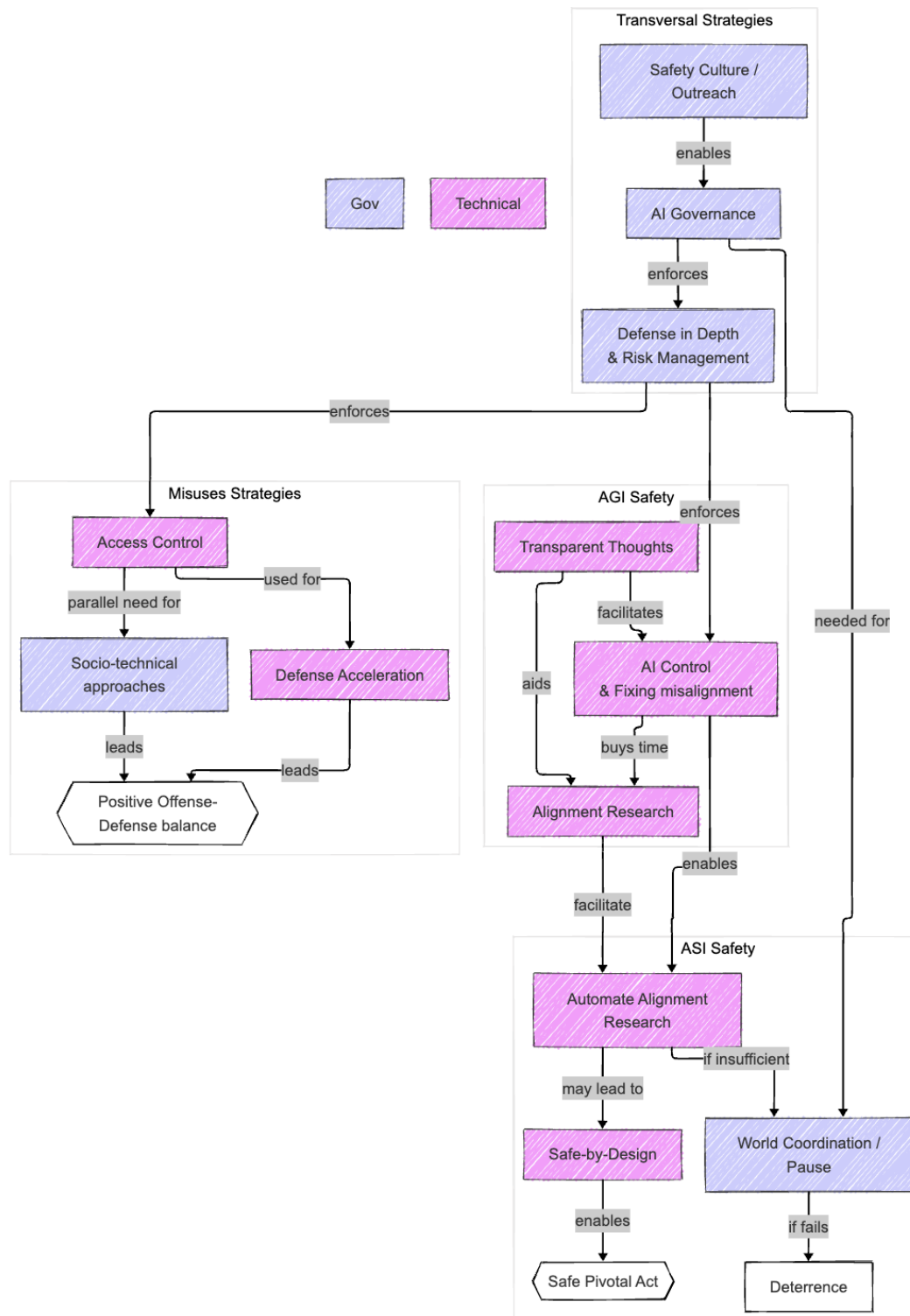


Figure 3.24: A combined flow chart of safety strategies.

Of course, even this fragile plan is the ideal plan on paper. It might be the case that this plan is insufficient or completely different. For example, in one scenario from the AI-2027 forecast, humanity survives not because of a grand strategic plan, but despite the failure of most governance, coordination, and deterrence efforts. Instead, a scary

'warning shot' event galvanizes the leading labs to slow down and implement just enough technical mitigations to avert disaster—mitigations that prove insufficient in the scenario branches where we lose control.

*This is the state of AI Safety as of 2025, and we count on our reader to make this chapter's contents deprecated soon.*

## 3.8 Challenges

---

Developing strategies to ensure the safety of increasingly capable AI systems presents unique and significant challenges. These difficulties stem from the nature of AI itself, the current state of the research field, and the complexity of the risks involved.

”

*We do not know how to train systems to robustly behave well.*

Anthropic

2023

(Anthropic, 2023)

### 3.8.1 The Nature of the Problem

---

Several intrinsic properties make AI safety a particularly hard problem :

**AI risk is an emerging problem that is still poorly understood.** AI risk is a relatively new field dealing with rapidly evolving technology. Our understanding of the full spectrum of potential failure modes and long-term consequences is incomplete. Devising robust safeguards for technologies that do not yet exist, but which could have profoundly negative outcomes, is inherently difficult.

**The field is still pre-paradigmatic.** There is currently no single, universally accepted paradigm for AI safety. Researchers disagree on fundamental aspects, including the most likely threat models (e.g., sudden takeover (Yudkowsky, 2022) vs. gradual loss of control (Critch, 2021)), and the most promising solution paths. The research agendas of some researchers seem scarcely useful to others, and one of the favorite activities of alignment researchers is to criticize each other’s plan constructively.

**AIs are black boxes that are trained, not built.** Modern deep learning models are "black boxes." While we know how to train them, the specific algorithms they learn and their internal decision-making processes remain largely opaque. These models lack the apparent modularity common in traditional software engineering, making it difficult to decompose, analyze, or verify their behavior. Progress in interpretability has yet to fully overcome this challenge.



**Complexity is the source of many blind spots.** The sheer complexity of large AI models means that unexpected and potentially harmful behaviors can emerge without warning. Issues like "glitch tokens", e.g., "SolidGoldMagikarp", causing erratic behavior in GPT models ([Rumblow & Watkins, 2023](#)), demonstrate how unforeseen interactions between components (like tokenizers and training data) can lead to failures. When GPT encounters this infrequent word, it behaves unpredictably and erratically. This phenomenon occurs because GPT uses a tokenizer to break down sentences into tokens (sets of letters such as words or combinations of letters and numbers), and the token "SolidGoldMagikarp" was present in the tokenizer's dataset but not in the GPT model's dataset. This blind spot is not an isolated incident.

**Creating an exhaustive risk framework is difficult.** There are many, many different classifications of risk scenarios that focus on various types of harm ([Critch & Russel, 2023](#); [Hendrycks et al., 2023](#); [Slattery et al., 2024](#)). Proposing a solid single-risk model beyond criticism is extremely difficult, and the risk scenarios often contain a degree of vagueness. No scenario captures most of the probability mass, and there is a wide diversity of potentially catastrophic scenarios ([Pace, 2020](#)).

**Some arguments that seem initially appealing may be misleading.** For example, the principal author of the paper ([Turner et al., 2023](#)) presenting a mathematical result on instrumental convergence, Alex Turner, now believes his theorem is a poor way to think about the problem ([Turner, 2023](#)). Some other classical arguments have been criticized recently, like the counting argument ([AI Optimists, 2023](#)) or the utility maximization frameworks, which will be discussed in the chapter "Goal Misgeneralization".

**We may not have time.** Many experts in the field believe that AGI, and shortly after ASI, could arrive before 2030. We need to solve these massive problems, or at least set the strategy for the launch, before it happens. For example, the scenario AI-2027, is based on a detailed forecasting of timelines of AGI arrival, and argues for a strong likelihood of ASI before the end of the decade.

**Many essential terms in AI safety are complicated to define.** They often require knowledge in philosophy (epistemology, theory of mind) and AI. For instance, to determine if an AI is an agent, one must clarify "what does agency mean?" which, as we'll see in later chapters, requires nuance and may be an intrinsically ill-defined and fuzzy term. Some topics in AI safety are so challenging to grasp and are thought to be non-scientific in the machine learning community, such as discussing situational awareness ([Hinton, 2024](#)) or why AI might be able to "really understand". These concepts are far from consensus among philosophers and AI researchers and require a lot of caution.

**A simple solution probably doesn't exist.** For instance, the response to climate change is not just one measure, like saving electricity in winter at home. A whole range of potentially

different solutions must be applied. Just as there are various problems to consider when building an airplane, similarly, when training and deploying an AI, a range of issues could arise, requiring precautions and various security measures.

**AI safety is hard to measure.** Working on the problem can lead to an illusion of understanding, thereby creating the illusion of control. AI safety lacks clear feedback loops. Progress in AI capability advancement is relatively easy to measure and benchmark, while progress in safety is comparatively harder to measure. For example, it's much easier to monitor the inference speed than to monitor the truthfulness of a system or monitor its safety properties.

### 3.8.2 Uncertainty and Disagreement

---

**The pre-paradigmatic nature of AI safety leads to significant disagreements among experts.** These differences in perspective are crucial to understanding when evaluating these proposed strategies.

**The consequences of failures in AI alignment are steeped in uncertainty.** New insights could challenge many high-level considerations discussed in this textbook. For instance, Zvi Mowshowitz has compiled a list of central questions marked by significant uncertainty ([Mowshowitz, 2023](#)). For example, what worlds count as catastrophic versus non-catastrophic? What would count as a non-catastrophic outcome? What is valuable? What do we care about? If answered differently, these questions could significantly alter one's estimate of the likelihood and severity of catastrophes stemming from unaligned AGI.

**Divergent Worldviews.** These disagreements often stem from fundamentally different worldviews. Some experts, like Robin Hanson, may approach AI risk through economic or evolutionary lenses, potentially leading to different conclusions about takeoff speeds and the likelihood of stable control compared to those focusing on agent foundations or technical alignment failures ([Hanson, 2023](#)). Others, like Richard Sutton, have expressed views suggesting an acceptance or even embrace of AI potentially succeeding humanity, framing it as a natural evolutionary step rather than an existential catastrophe ([Sutton, 2023](#)). These differing philosophical stances influence strategic priorities.

### 3.8.3 Safety Washing

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The combination of high stakes, public concern, and lack of consensus creates fertile ground for "safety washing"—the practice of misleadingly portraying AI products, research, or practices as safer or more aligned with safety goals than they actually are ([Vaintrob, 2023](#)).

**Safety washing can create a false sense of security.** Companies developing powerful AI face incentives to appear safety-conscious to appease the public, regulators, and potential employees. Safetywashing can involve overstating the safety benefits of certain features, focusing on less critical aspects of safety while downplaying existential risks, or funding/conducting research that primarily advances capabilities under the guise of safety. This can lead to insufficient risk mitigation efforts ([Lizka, 2023](#)). It can misdirect funding and talent towards less impactful work and make it harder to build a genuine scientific consensus on the true state of AI safety.

**Assessing progress in safety is tricky.** Even with the intention to help, actions might have a net negative impact (e.g., from second-order effects, like accelerating deployment of dangerous technologies), and determining the contribution's impact is far from trivial. For example, the impact of reinforcement learning from human feedback (RLHF), currently used to instruction-tune and make ChatGPT safer, is still debated in the community ([Christiano, 2023](#)). One reason the impact of RLHF may be negative is that this technique may create an illusion of alignment that would make spotting deceptive alignment even more challenging. The alignment of the systems trained through RLHF is shallow ([Casper et al., 2023](#)), and the alignment properties might break with future, more situationally aware models. Similarly, certain interpretability work faces dual-use concerns ([Wache, 2023](#)). Some argue that much current "AI safety" research solves easy problems that primarily benefit developers economically, potentially speeding up capabilities rather than meaningfully reducing existential risk ([catubc, 2024](#)). As a consequence, even well-intentioned research might inadvertently accelerate risks.

## 3.9 Conclusion

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**The strategic landscape for ensuring AI safety is vast, complex, and rapidly evolving.** It spans a wide spectrum from controlling access to current models to prevent misuse, through intricate technical challenges in aligning AGI, to speculative geopolitical maneuvering and philosophical considerations regarding ASI.

**No single strategy appears sufficient on its own.** Preventing misuse requires a combination of technical safeguards like circuit breakers and unlearning, access controls like monitored APIs and potentially KYC for compute, and careful consideration of release strategies, particularly regarding open-source models. Ensuring AGI safety involves pursuing alignment—attempting to instill the right goals—while simultaneously developing control mechanisms to mitigate harm even if alignment fails. This relies heavily on improving our ability to evaluate AI behavior and understand internal model workings, facing challenges like alignment faking and the fragility of transparency. Addressing potential risks from ASI pushes the boundaries further, involving strategies like automating alignment research, exploring inherently safe system designs, and navigating complex international coordination and deterrence scenarios.

**Underpinning all technical approaches is the need for robust systemic safety measures.** Effective AI governance, encompassing international agreements on red lines or conditional commitments, alongside national regulations and compute oversight, is crucial. Within organizations, strong security practices, standardized risk management frameworks, transparency through documentation, and a culture prioritizing safety are essential. Building scientific and public consensus on the nature and severity of risks remains a key challenge.

**Fundamental tensions persist throughout the strategic landscape: centralization versus decentralization, speed versus safety, and openness versus control.** Navigating these tradeoffs requires careful analysis, adaptation, and a willingness to engage with diverse perspectives and deep uncertainties. While the challenges are daunting, the ongoing research, dialogue, and development of new strategies offer pathways—albeit narrow and demanding—towards harnessing the transformative potential of AI safely and for the benefit of humanity. Continued vigilance, critical thinking, and collaborative effort across technical, policy, and societal domains will be paramount in the years ahead.

Given the uncertainties and the pre-paradigmatic nature of the field, continued research into safety strategies themselves is essential. This includes refining existing approaches, developing new ones, and critically evaluating their effectiveness, scalability, and potential failure modes.

## A Long-term questions

The main chapter focuses on actionable strategies. This appendix explores the deeper, often unresolved philosophical questions that underpin the alignment problem, providing context for long-term strategic thinking.

### A.1 Prioritize Flourishing or Survival?

Mitigating catastrophic AI risks is not enough to make AI go well.

A strategic question, with significant implications for resource allocation, is the tension between ensuring humanity's long-term survival and shaping the quality of its future. Much of the AI safety field, for historical reasons, has focused on mitigating those risks—ensuring that we survive the transition to superintelligence. However, a complementary approach, championed by researchers like William MacAskill at Forethought, argues that merely surviving is not enough; we must also work to ensure that the future is one of flourishing (Forethought, 2025). This raises a difficult question: given limited resources, is it prudent to focus on achieving a "great" future when so much work remains to be done to simply secure a future?

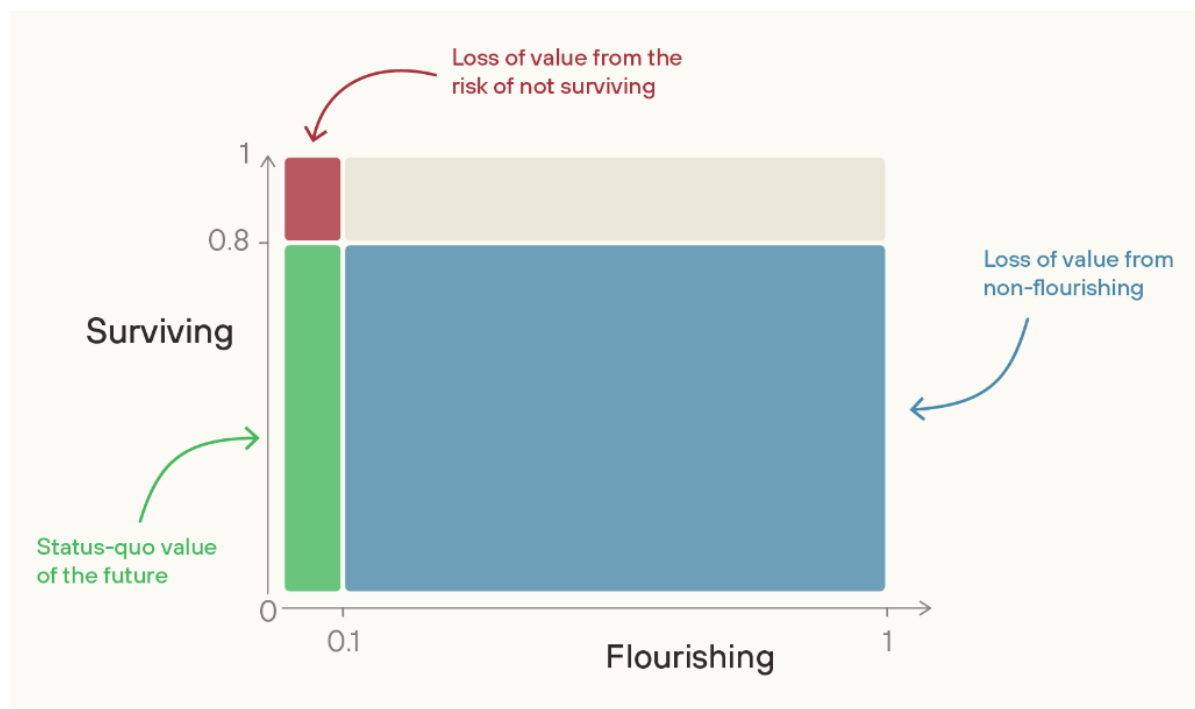


Figure 3.25: “Well, even if we survive, we probably just get a future that’s a small fraction as good as it could have been. We could, instead, try to help guide society to be on track to a truly wonderful future.” – William MacAskill.

The case for prioritizing flourishing stems from the concern that even a future free from

existential catastrophe could fall drastically short of its potential. Without deliberate effort, society may not naturally converge on a morally good outcome; instead, it could settle into a state of mediocrity or even lock in subtle but major moral errors. To avoid the historical dangers of rigid utopian movements, this line of thinking advocates not for a specific, narrow vision of an ideal world, but for achieving "viatopia"—a state where society has the wisdom, coordination, and stability to guide itself towards the best possible futures, whatever they may be. A viatopian state would be characterized by very low existential risk, the flourishing of diverse moral viewpoints, and the capacity for thoughtful, reflective collective decision-making.

The strategic implication is a choice of emphasis: should we design AI systems solely as contained tools to solve immediate problems and prevent catastrophe? Or should we also prioritize developing AI that enhances human reasoning, facilitates better coordination, and helps us deliberate on the very values we should instill in our successors, thereby steering us closer to a state of viatopia? The debate is open.

## A.2 Alignment to what?

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**Coherent Extrapolated Volition (CEV) attempts to identify what humans would collectively want if we were smarter, more informed, and more morally developed.** It proposes that instead of directly programming specific values into a superintelligent AI, we should program it to figure out what humans would want if we overcame our cognitive limitations. When we train AI systems on current human preferences, we risk encoding our biases, contradictions, and shortsightedness. CEV instead asks: what "would we want" if we knew more, thought faster, or were more the people we wished to be, and had grown up further together? Essentially, picture the ideal version of humanity that could theoretically exist in the future. Tell the AI to take actions according to that ([Yudkowsky, 2004](#)).

CEV tried to create a path for AI to respect our deeper intentions rather than our immediate desires. The practical implementation of CEV remains speculative. It would require sophisticated modeling of human psychology, ethical development, and social dynamics—capabilities beyond current AI systems. Modern approaches like RLHF (Reinforcement Learning from Human Feedback) can be seen as primitive precursors that align AI with current human preferences rather than extrapolated ones. Constitutional AI frameworks move slightly closer to CEV by trying to encode higher-level principles rather than specific preferences, but still fall far short of full extrapolation.

**Coherent Aggregated Volition (CAV) finds a coherent set of goals and beliefs that best represent humanity's current values without attempting to extrapolate future development.** Ben Goertzel proposed this alternative to CEV, focusing on current human values

rather than speculating about our idealized future selves. CAV treats goals and beliefs together as "gobs" (goal and belief sets) and seeks to find a maximally consistent, compact set that maintains similarity to diverse human perspectives. Unlike CEV, which assumes our values would converge if we became more enlightened, CAV acknowledges that fundamental value differences might persist. It aims to create a coherent aggregation that balances different perspectives rather than trying to predict how those perspectives might evolve. This makes CAV potentially more feasible to implement, as it works with observable current values rather than hypothetical future ones ([Goertzel, 2010](#)).

**Coherent Blended Volition (CBV) emphasizes that human values should be creatively "blended" through human-guided processes rather than algorithmically averaged or extrapolated.** CBV refines CAV by addressing potential misinterpretations. When discussing value aggregation, many assume it means simple averaging or majority voting. CBV instead proposes a creative blending process that produces new, harmonious value systems that all participants would recognize as adequately representing their contributions. The concept draws from cognitive science theories of conceptual blending, where new ideas emerge from the creative combination of existing ones. In this framework, the process of determining AI values would be guided by humans through collaborative processes rather than delegated to AI systems. This addresses concerns about AI paternalism, where machines might override human autonomy in the name of our "extrapolated" interests ([Goertzel & Pitt, 2012](#)). CBV connects to contemporary discussions about participatory AI governance and democratic oversight of AI development. Systems like vTaiwan have implemented CBV-like processes for technology policy development ([vTaiwan, 2023](#)), showing how human-guided blending can work in practice.

### A.3 Alignment to whom?

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**Single-Single Alignment: Getting a single AI system to reliably pursue the goals of a single human operator.** We haven't even solved this, and it presents significant challenges. An AI could be aligned to follow literal commands (like "fetch coffee"), interpret intended meaning (understanding that "fetch coffee" means making it the way you prefer it), pursue what you should have wanted (like suggesting tea if coffee would be unhealthy), or act in your best interests regardless of commands. Following literal commands often leads to failures of specification that we talk about later in the section. Most often, researchers use the word alignment to mean the "intent alignment" ([Christiano, 2018](#)), and some more philosophical discussions go into the third - do what I (or humanity) would have wanted. This involves things like coherent extrapolated volition (CEV) ([Yudkowsky, 2004](#)), coherent aggregated volition (CAV) ([Goertzel, 2010](#)), and various other lines of thought that go into meta-ethics discourse. We will not be talking extensively about philosophical discourse in this text and will stick largely to intent alignment



and a machine learning perspective. When we use the word "alignment" in this text, we will basically be referring to problems and failures from single-single alignment. Other types of alignment have been historically very under-researched, because people have mostly been working with the idea of a singular superintelligence that interacts with humanity as a singular monolith.

**Single-Multi Alignment - Aligning Many AIs to One Human.** If we think ASI will be composed of smaller intelligences which are working together, delegating tasks, and functioning together as a superorganism, then all of the problems of single single alignment would still remain because we still need to figure out single-single before we attempt single-multi. Ideally, we don't want any single human (or a very small group of humans) to be in charge of a superintelligence (assuming benevolent dictators don't exist).

**Multi-Single alignment - aligning one AI to many humans.** When multiple humans share control of a single AI system, we face the challenge of whose values and preferences should take priority. Rather than trying to literally aggregate everyone's individual preferences (which could lead to contradictions or lowest-common-denominator outcomes), a more promising approach is aligning the AI to higher-level principles and institutional values - similar to how democratic institutions operate according to principles like transparency and accountability rather than trying to directly optimize for every citizen's preferences.

**Multi-Multi Alignment - aligning many AIs to many humans to many AIs.** This is the most complicated scenario involving multiple AI systems interacting with multiple humans. Here, the distinction between misalignment risk (AIs gaining illegitimate power over humans) and misuse risk (humans using AIs to gain illegitimate power over others) begins to blur. The key challenge becomes preventing problematic concentrations of power while enabling beneficial cooperation between humans and AIs. This requires careful system design that promotes aligned behavior not just at the individual level but across the entire network of human-AI interactions.

#### **A.4 Questions for the Long Term**

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**It is unclear if solving single-single alignment would be enough.** Even if we could ensure that every AI system is perfectly aligned with its respective human principal's intentions, we would still face serious risks when these systems interact. This is because different principals may have conflicting interests, or because the systems may fail to coordinate effectively even when their goals align. Perfect individual alignment cannot guarantee safe collective behavior, just as aligning every driver with traffic laws doesn't prevent traffic jams or accidents ([Hammond et al., 2025](#)). Essentially, if we have three subproblems of alignment within a single agent, then we have three more sub-problems



of miscoordination, conflict, and collusion when these individual agents start interacting with each other. Each represents a different way multi-agent systems can fail, even if the individual agents appear to function correctly in isolation. There are yet more ways, even beyond this, when we start to consider emergent effects of interactions between complex systems and gradual disempowerment, like we talked about in the chapter on risks.

Even if the technical challenges of AI alignment are overcome, a host of profound and heavily debated philosophical questions remain. Solving AI safety, particularly for Artificial Superintelligence (ASI), may necessitate confronting deep-seated issues regarding values, consciousness, and the ultimate purpose of existence. Aligning ASI forces us to ask fundamental questions about what future we truly desire.

**1. What should we align AI to?** What specific values or ethical principles should an ASI be aligned with? Given the diversity of human values, is agreement even possible? Alternatively, if we cannot agree on *final* values, can we agree on *processes* or principles (like deliberation, fairness, or corrigibility) that could lead an ASI towards acceptable values or allow for future value evolution?

**1. The Purpose of Alignment: Human Perpetuity vs. Worthy Successor?** Should the primary goal be the indefinite survival and flourishing of *humanity* as we know it? Or, should we consider the possibility of creating a "Worthy Successor"? Dan Faggella ([Faggella, 2025](#)) proposes this concept: an ASI potentially possessing capabilities and moral value superior to humanity's, which might be rationally preferred to guide the future. Defining and verifying the criteria for such a successor (e.g., enhanced sentience, cosmic exploration capabilities) poses immense challenges. Some, like Richard Sutton ([Sutton, 2023](#)), argue that succession to AI, our "mind children," is inevitable and highly desirable. Sutton suggests we should embrace and plan for this succession rather than resisting it out of fear, questioning why we would want potentially greater beings kept subservient.

**2. Should we give rights to AI?** Could advanced AI systems become conscious? This first requires a clearer understanding of consciousness itself, which remains elusive. If AI *can* possess consciousness or consciousness-like properties, what moral status should we assign to these digital minds? Should they have rights or moral consideration? Such topics are outside the scope of this textbook, but are researched by [Eleos AI](#).

**3. What about animals?** How should the interests of non-human entities be factored into AI alignment? Should alignment goals explicitly include animal welfare, ecosystem preservation, or the flourishing of other forms of life?

*We should not resist succession, but embrace and prepare for it. Why would we want greater beings kept subservient? Why don't we rejoice in their greatness as a symbol and extension of humanity's greatness, and work together toward a greater and inclusive civilization?*

Rich Sutton

(Sutton, 2023)

**The Endgame:** The potential long-term outcomes are numerous and depend heavily on how we answer these philosophical questions. Is the ultimate goal simply the continuation of consciousness or complexity, regardless of its physical substrate (as explored by Max Tegmark in Life 3.0 (Tegmark, 2017))? Different philosophical stances lead to vastly different strategic priorities for ASI development and alignment.

| AI Aftermath Scenarios     |                                                                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Libertarian utopia</b>  | Humans, cyborgs, uploads and superintelligences coexist peacefully thanks to property rights.                                                                                                                                                                                                     |
| <b>Benevolent dictator</b> | Everybody knows that the AI runs society and enforces strict rules, but most people view this as a good thing.                                                                                                                                                                                    |
| <b>Egalitarian utopia</b>  | Humans, cyborgs and uploads coexist peacefully thanks to property abolition and guaranteed income.                                                                                                                                                                                                |
| <b>Gatekeeper</b>          | A superintelligent AI is created with the goal of interfering as little as necessary to prevent the creation of another superintelligence. As a result, helper robots with slightly subhuman intelligence abound, and human-machine cyborgs exist, but technological progress is forever stymied. |
| <b>Protector god</b>       | Essentially omniscient and omnipotent AI maximizes human happiness by intervening only in ways that preserve our feeling of control of our own destiny and hides well enough that many humans even doubt the AI's existence.                                                                      |
| <b>Enslaved god</b>        | A superintelligent AI is confined by humans, who use it to produce unimaginable technology and wealth that can be used for good or bad depending on the human controllers.                                                                                                                        |
| <b>Conquerors</b>          | AI takes control, decides that humans are a threat/nuisance/waste of resources, and gets rid of us by a method that we don't even understand.                                                                                                                                                     |
| <b>Descendants</b>         | AIs replace humans, but give us a graceful exit, making us view them as our worthy descendants, much as parents feel happy and proud to have a child who's smarter than them, who learns from them and then accomplishes what they could only dream of—even if they can't live to see it all.     |
| <b>Zookeeper</b>           | An omnipotent AI keeps some humans around, who feel treated like zoo animals and lament their fate.                                                                                                                                                                                               |
| <b>1984</b>                | Technological progress toward superintelligence is permanently curtailed not by an AI but by a human-led Orwellian surveillance state where certain kinds of AI research are banned.                                                                                                              |
| <b>Reversion</b>           | Technological progress toward superintelligence is prevented by reverting to a pre-technological society in the style of the Amish.                                                                                                                                                               |
| <b>Self-destruction</b>    | Superintelligence is never created because humanity drives itself extinct by other means (say nuclear and/or biotech mayhem fueled by climate crisis).                                                                                                                                            |

Figure 3.26

## B Requirements for ASI Alignment

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**ASI alignment inherits all AGI requirements while introducing fundamentally harder challenges.** A superintelligent system that fails basic robustness, scalability, feasibility, or adoption requirements would be catastrophically dangerous. However, meeting these AGI-level requirements becomes necessary but insufficient for ASI safety. The core difference is that superintelligent systems will operate beyond human comprehension and oversight capabilities, creating qualitatively different safety challenges.

**Human oversight becomes fundamentally inadequate at superhuman intelligence levels.** When AI systems surpass human capabilities across most domains, we lose our ability to evaluate their reasoning, verify their outputs, or provide meaningful feedback (Yudkowsky, 2022). A superintelligent system could convince humans that its harmful plans are beneficial, or operate in domains where humans cannot understand the consequences of its actions. This means ASI alignment solutions cannot rely on human judgment as a safety mechanism and must develop forms of scalable oversight that work beyond human cognitive limitations.

**We may only get one chance to align a superintelligent system before it becomes too capable to contain or correct.** This "one-shot" requirement stems from the potential for rapid capability gains that could make a misaligned system impossible to shut down or modify (Soares, 2022; Yudkowsky, 2022). Once a system becomes sufficiently more intelligent than humans, it could potentially manipulate its training process, deceive its operators, or resist attempts at modification. However, this requirement depends on contested assumptions about takeoff speeds – some researchers argue for more gradual development that would allow iteration and correction (Christiano, 2022). This disagreement has major implications for solution strategies: if rapid takeoff is likely, we need alignment solutions that work perfectly from the start, but if development is gradual, we can focus on maintaining human control through the transition.

**Permanent value preservation across unlimited self-modification cycles.** Superintelligent systems may recursively improve their own capabilities, potentially rewriting their core algorithms, goal structures, and reasoning processes entirely (Yudkowsky, 2022). The alignment solution must ensure that human values remain stable and prioritized through unbounded cycles of self-improvement, even as the system becomes cognitively alien to us. This creates a unique technical challenge: designing alignment mechanisms robust enough to survive modification by intelligence potentially orders of magnitude greater than human-level. Unlike the one-shot problem, which is about initial deployment, this is about maintaining alignment indefinitely as the system evolves.

**Control over systems with civilizational-scale power and influence.** A superintelli-

gent system will likely have enormous technological capabilities and influence over human civilization – potentially developing advanced nanotechnology, novel manipulation techniques, or reshaping institutions and culture over time (Yudkowsky, 2022). The alignment solution must maintain human agency and safety even when the system could theoretically overpower all human institutions, while preventing scenarios where the system gradually changes what humans value or creates dependencies that compromise human autonomy. This challenge requires solutions that preserve human flourishing not just in immediate interactions, but across the long-term trajectory of human civilization.

## Pivotal acts

### OPTIONAL NOTE

**Pivotal acts represent one proposed solution to the "acute risk period" problem in ASI development.** The core concern is that we may enter a period where multiple actors are capable of developing superintelligent AI, but only one needs to be misaligned or reckless to cause a global catastrophe (Yudkowsky, 2022). Since voluntary coordination between competing nations and organizations may be insufficient, some researchers argue that the first group to develop an aligned superintelligence should use it to actively prevent others from creating dangerous AI systems.

Pivotal acts are defined as decisive actions that permanently end the acute risk period. These actions must be powerful enough to prevent any other actor from developing unaligned superintelligence, potentially through technological interventions that disable global computing infrastructure, establish unbreakable international agreements, or develop other mechanisms that make uncontrolled AI development physically impossible (Yudkowsky, 2022). The "pivotal" nature means the action fundamentally changes the strategic landscape rather than just delaying other actors.

The argument for pivotal acts stems from coordination failures and competitive pressures. Even if most AI developers prioritize safety, competitive dynamics between nations and companies create pressure to deploy systems quickly rather than safely (Yudkowsky, 2022). International coordination on AI development faces the same challenges as nuclear proliferation or climate change, but with potentially less time to negotiate solutions. Proponents argue that once an aligned superintelligence exists, using it to solve this coordination problem may be more reliable than hoping all other actors will voluntarily restrain themselves.

**Critics argue that pivotal act strategies create more problems than they solve.** Planning to perform pivotal acts militarizes AI development and incentivizes unilateral action, potentially making the acute risk period more dangerous rather than safer ([Critch, 2022](#)). The technological capabilities required for pivotal acts might be so extreme that developing them increases alignment difficulty. Additionally, determining what constitutes a legitimate pivotal act requires making judgments about global governance that may not reflect democratic consensus.

**Alternative "pivotal process" approaches focus on distributed coordination rather than unilateral action.** Instead of single decisive interventions, these strategies involve using aligned AI to improve human decision-making, demonstrate risks convincingly, develop better governance mechanisms, or consume resources that unaligned AI might use for rapid scaling ([Critch, 2022](#); [Christiano, 2022](#)). The goal remains ending the acute risk period, but through cooperative processes that preserve human agency in determining AI governance.

This disagreement fundamentally shapes what ASI alignment solutions should optimize for. Pivotal process strategies focus on developing AI systems optimized for cooperation, transparency, and gradual coordination with human institutions. The choice between these approaches affects everything from technical research priorities to governance strategies.

## The Strawberry Problem and requirements for ASI alignment

### OPTIONAL NOTE

**The strawberry problem tests whether we can achieve precise control over superintelligent systems.** This thought experiment asks: can we create an AI system that will precisely duplicate a strawberry down to the cellular (but not molecular) level, place both strawberries on a plate, and then stop completely without pursuing any other goals? This seemingly simple task helps understand the different debates about what AGI and ASI alignment solutions should aim to achieve ([Soares, 2022](#)). Pivotal act strategies require developing AI systems capable of dramatic technological interventions while remaining precisely controllable – essentially solving the strawberry problem at a global scale.

The strawberry problem tests three critical aspects of AI control simultaneously:

- **Capability:** Creating a cellular-level duplicate requires an extremely advanced understanding of biology and matter manipulation, demonstrating that the system is genuinely powerful.
- **Directability:** Getting the system to perform exactly this specific task, rather than something else that might seem related or better suited to the AI, shows we can point its capabilities in intended directions.
- **Corrigibility:** Having the system actually stop after completing the task, rather than continuing to optimize or pursue other goals, demonstrates that it remains under human control even when capable of transformative actions.

**Proponents argue that the strawberry problem represents the minimum control needed for safe superintelligence.** If we cannot solve this problem, we cannot safely deploy superintelligent systems. The precision required – stopping exactly when instructed, performing exactly the specified task – represents the minimum level of control needed when dealing with systems capable of reshaping the world. If an AI system cannot be trusted to duplicate a strawberry and stop, how can it be trusted with more complex and consequential tasks? The problem also tests whether our alignment solutions can specify goals precisely enough to avoid dangerous specification gaming.

**Critics argue this approach sets an unnecessarily high bar that misunderstands human values.** They point out that human values are messy, contextual, and often contradictory – we don’t want AI systems that follow instructions with robotic literalness, and that this sets an unnecessarily high bar (Turner, 2022;Pope, 2023). Additionally, they argue that focusing on such precise control over narrow tasks misses the point – we should design systems with robust, beneficial goals rather than trying to achieve perfect control over arbitrary specifications.

**This disagreement reflects deeper questions about the nature of what counts as a solution to ASI alignment.** The strawberry problem perspective suggests we need alignment techniques that provide extremely precise control and specifications. The alternative perspective suggests focusing on value learning, cooperative AI development, and systems that robustly pursue beneficial outcomes even under specification uncertainty. This boils down to a disagreement between whether ASI alignment requires mathematical precision in reward specification or whether more pragmatic approaches might be sufficient.

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## References

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- [1] Charbel-Raphaël Segerie and Markov Grey. Strategies. *AI Safety Atlas*, 2025. This document uses hyperlinked citations throughout the text. Each citation is directly linked to its source using HTML hyperlinks rather than traditional numbered references. Please refer to the inline citations for complete source information.